How Do Underrepresented Voters View Electoral System Trade-Offs?

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Abstract

Scholars of electoral reform have long debated the trade-off between accountability and representation in majoritarian versus proportional systems, yet few have examined how voters themselves view this trade-off. We argue that voters who have historically lacked political voice, such as women and young people, are more likely to favor reforms that expand representation, even at some cost to accountability. We test this argument using a conjoint experiment conducted before South Korea's 2020 election—a moment marked by intense public debate over whether the government should implement electoral reforms to enhance representation or accountability. Our findings show that while women and younger voters lean toward more representative rules, they also value accountability and resist certain reforms that would weaken it. This preference for representation is stronger when considering major electoral system changes rather than minor rule adjustments and is especially pronounced among independent or minor party supporters compared to major party supporters. These insights have important implications for our understanding of electoral reform, voter preferences, and the political inclusion of underrepresented groups.

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One of the most active debates in political science concerns the consequences of electoral systems. The extensive literature on electoral institutions suggests that different electoral systems can lead to varying outcomes, representing a trade-off between distinct political goals. Majoritarian rules are often viewed as providing greater accountability, while proportional rules are seen as favoring broader representation (Cox 1997; Powell 2000; Shugart and Carey 1992). For this reason, scholars of electoral reform have debated which electoral system provides the optimal balance, aiming to maximize both representation and accountability without excessively compromising either dimension (Carey and Hix 2011; Persson and Tabellini 2003).

Within this large literature on electoral consequences, less attention has been given to how voters themselves respond to changes in electoral rules. In this article, we examine how electoral reform shapes individual voters' preferences over electoral systems, focusing in particular on how voters view the trade-off between accountability and representation. We theorize that voters who have historically lacked political voice—such as women and young people—will be more supportive of proportional rules that increase representation over majoritarian rules that emphasize accountability.

Research on public support for electoral systems is not entirely new. Scholars of electoral politics have compared citizens' overall satisfaction in democratic countries with majoritarian versus proportional electoral rules (Aarts and Thomassen 2008; Anderson and Guillory 1997; Birch 2008; Lijphart 2012). Over time, the lens through which voter preferences toward electoral institutions are understood has evolved (see Blais et al. 2021; Blais, Sevi, and Plescia 2022). Early work relied on cross-national analyses. Later observational studies examined actual referendums on electoral reform—moves toward either more majoritarian or more proportional rules (Banducci and Karp 1999; Riambau, Steven, and Boe-Gibson 2021; Wenzel, Bowler, and Lanoue 2000). These studies, however, typically captured aggregate public opinion shaped by contemporary partisan factors, rather than the underlying preferences of specific groups of voters. More recently, survey experiments have been used

to test voter preferences under hypothetical electoral conditions (Hoffmann and Renes 2022; Linhart, Jankowski, and Tepe 2023; Nagtzaam and van Erkel 2017; Plescia, Blais, and Högström 2020). While these studies provide valuable insights, they tend to rely on simplified scenarios, leaving open questions about how voters respond to specific reforms when they are embedded in an ongoing political debate.

We build on this research by situating our study in the context of a real-world electoral reform. Specifically, we conducted a conjoint experiment immediately before South Korea's 2020 general election. This timing allowed us to capture voter preferences during a period of intense public debate over whether the government should revise the current electoral rules to strengthen representation or accountability. As detailed below, after its 1988 democratization South Korea used a pure mixed-member majoritarian (MMM) system that placed a greater emphasis on accountability for over three decades. Following the 2020 general election, however, the country shifted toward a more representative electoral system that combined MMM and mixed-member proportional (MMP) rules.

Our findings show that while women and younger respondents were more likely than men and older respondents to endorse rules that prioritize representation, this support had important limits, indicating that these groups also value accountability. For example, women were more supportive than men of the proposed combination of MMM and MMP—which would enhance representation over the pre-reform MMM-only system—but they did not favor a full transition to MMP, the system most tilted toward representation at the expense of accountability. Similarly, younger respondents expressed stronger support for MMP than older respondents but were reluctant to weaken accountability in other areas, such as by reducing the number of majoritarian seats to expand the PR tier. Moreover, we also find that preferences for representation were stronger for the overall choice of electoral system (e.g., MMM versus MMP) than for more technical rule changes, such as vote aggregation methods in the PR tier.

Importantly, these preferences are not explained solely by partisan differences. Even

among supporters of the major liberal party, whose president openly advocated for the more representative MMM-MMP mix, women were more likely than men to support this shift. Gender and age differences were also especially pronounced among independents and supporters of smaller parties. For these groups, their dual underrepresentation—stemming from both their party affiliation and demographic characteristics—may heighten their preference for more representative electoral rules.

To the best of our knowledge, this is the first study to use conjoint survey experiments on electoral reform to directly examine how voters perceive the representation-accountability trade-off. Our approach builds on and complements prior work in several ways. First, our conjoint design generates a rich set of counterfactuals by randomly combining attributes of electoral institutions, extending beyond the cross-national and referendum-based studies that have dominated the literature. Second, whereas earlier studies often examined overall public opinion, we disaggregate preferences by key demographic characteristics such as gender and age. By randomizing electoral rules across alternatives supported by different parties, our results also mitigate concerns that findings simply reflect partisan alignments. Third, by situating the experiment in the midst of an actual electoral reform debate, we provide a higher degree of contextual realism than survey experiments that rely on more simplified scenarios. Taken together, these features allow us to offer new insights into how voters—particularly those from underrepresented groups—evaluate the trade-offs between representation and accountability in the context of electoral reform.

Representation vs. Accountability

The importance of electoral systems lies in the fact that the basic form of democracy—procedural democracy or electoral democracy—requires "free and fair elections" as a fundamental element (Dahl 1989). Once free and fair elections are achieved through democratization, the natural process of political development in democracies involves identifying problems within the initial electoral system and discussing how to improve it via electoral

reform for the consolidation of democracy. Among the various elements that characterize electoral systems, central to the discussion of electoral reform is how to translate votes to seats—the method of determining a winner from the outcome of the vote (Taagepera and Shugart 1989). Across a range of electoral systems and voting rules, there is a majoritarian type of winner-take-all rule at one end of the continuum, adopted in national elections in the United States and United Kingdom. At the other end, there is a proportional style of representation rule, used in many European consensus democracies (Lijphart 2012).

In this article, we examine how voters view the balance between accountability and representation, an important trade-off inherent in these electoral rules (Shugart and Carey 1992). Majoritarian systems tend to simplify accountability by limiting the number of viable parties, thereby making it easier for voters to identify governing alternatives and hold them responsible at the ballot box. Proportional systems, in contrast, emphasize representation by lowering barriers for smaller parties and enabling a wider range of social groups to enter the legislature. We focus on representation primarily in its descriptive sense—who gains seats and has a voice in the assembly—while acknowledging that descriptive gains can foster substantive representation, as more diverse legislators may be better positioned to advance the interests of historically underrepresented groups.

The trajectory of electoral institutions varies across countries, but given the observation of the third wave of democratization in Asia, during the early stages of democracy, many new democracies often adopt majoritarian systems (presidentialism combined with plurality electoral rules), emphasizing efficiency and simplifying state governance. This was the case in the Philippines, Taiwan, and South Korea. However, as states and governments gradually stabilize, democratic countries often consider transforming their political institutions from those that facilitate the concentration of power to ones that incorporate the interests of diverse groups in society, especially the voices of the underprivileged and minorities, into a proportional system (e.g., Batto et al. 2016). As argued by Lijphart (2012), power-concentrating political institutions, such as majoritarian electoral rules, have the nature of

a zero-sum game and can cause tensions and conflicts between social groups. Therefore, as society becomes more diverse, power-dispersing political institutions, such as proportional representation electoral rules, are often considered as more appropriate and necessary for the consolidation of (consensus) democracy (Reynolds and Reilly 1999).

Yet, it is not entirely clear which type of electoral system citizens perceive as accurately reflecting their preferences. Proportional representation is commonly regarded as a fairer system for translating votes into seats than majority or plurality rule, making it more appealing to those who care about justice in processes or outcomes (Gangl 2003; Miller 2001). Some studies show that more proportional systems positively affect voters' perception of electoral fairness (Karp and Banducci 2008) and general satisfaction with the system (Banducci, Donovan, and Karp 1999; Lijphart 2012; Plescia et al. 2020). Conversely, individuals also value government efficiency and may prefer the clearer accountability that majority or plurality rule provides through decisive electoral outcomes. Since citizens often want stronger and steadier governments rather than unstable ones (Gallgher et al. 2011; Warwick 1994), which may fail to meet their electoral promises and election pledges (Strøm and Swindle 2002; Thomson et al. 2017), it is unsurprising that some studies find higher levels of overall satisfaction and perceived electoral integrity in democracies with disproportional rather than proportional electoral systems (Aarts and Thomassen 2008; Norris 1999).

Scholars have investigated the reasons for these seemingly contradictory findings, paying particular attention to research designs and contextual factors. Some inconsistencies in the findings may be related to the reliance on cross-national data and the failure to isolate the causal effect of electoral institutions. Other discrepancies could arise from the influence of contemporary political aspects related to electoral reform and the dependence on incomplete voting conditions (see Blais et al. 2021; Blais et al. 2022). Recent studies have adopted experimental designs to address these issues and provide stronger causal inferences. However, much of the experimental research tends to view the effects of electoral institutions on voter preferences as conditional on individuals' partisan considerations (e.g., Blais et al. 2022;

Plescia et al. 2020). While we acknowledge that partisan attachments play a significant role in voters' assessments of electoral rules, we argue that these assessments are also influenced by political status based on demographic characteristics. This is especially true for voter groups, such as women and young people, who have not been properly represented by political parties within the current system and tend to be relatively disadvantaged in the political arena. These groups may see themselves as better represented under a new system through electoral reform. In the next section, we introduce several fundamental reasons why these voter groups are likely to prefer proportional rules to majoritarian ones.

Which Rules Do Underrepresented Voters Prefer?

We expect that voters who have historically lacked political voice will favor more proportional voting rules that expand representation over majoritarian rules that emphasize accountability.[1] This underrepresentation has often resulted in policies that overlook these groups' needs and concerns, contributing to a lack of trust in the political system. While these groups also value accountability, we expect them to prioritize electoral rules that correct this imbalance and give them a stronger voice in policymaking. More specifically, we focus on young and female voters, arguing that their preference for proportional rules stems from both value-based and self-interested motivations (Bol et al. 2023).

First, underrepresented voters may favor proportional rules because these rules can create a fairer, more inclusive, and more representative political system. Proportional systems allocate legislative seats in proportion to the number of votes each party or candidate receives, in contrast to the winner-take-all, zero-sum nature of majoritarian systems (Taagepera and Shugart 1989). This means that parties or candidates who represent minority groups can win seats and have a voice in policymaking, even if they do not win a majority (or plurality) of votes. Proportional rules may also increase trust in government by reducing the sense that some votes are wasted (Banducci et al. 1999). In majoritarian systems, votes for losing

[1] We pre-registered our main expectations at Open Science Framework (OSF) before data analysis. Further details on our pre-registration plan and additional analyses can be found in the Appendix.

candidates or parties have no impact on seat allocations, which can lead to a significant portion of the electorate feeling unrepresented (Lijphart 1984). In contrast, proportional rules ensure that every vote counts toward the allocation of seats, giving voters more confidence that their voice is being heard (Guinier 1994).

Second, underrepresented voters may also value proportional rules because they can increase the diversity of parties in the legislature. Under majoritarian rules, major parties often have a stranglehold on power, making it difficult for newer or smaller parties to gain a foothold. In comparison, proportional rules allow for greater competition and give smaller parties a better chance at winning seats (see Shugart and Wattenberg 2001). This can result in a more diverse range of parties and perspectives being represented in government, including those that may represent minority communities. Moreover, while major parties typically dominate the legislative process in majoritarian systems, proportional systems often require parties to form coalitions or work together to pass legislation (e.g., Strøm, Müller, and Bergman 2008). This can lead to a more collaborative and compromise-driven political system and culture (Lijphart 2012), which can be beneficial for politically underrepresented groups who may have less influence on their own.

Finally, underrepresented voters may favor proportional rules because they can increase the chances that members of these groups will be elected as representatives themselves. In majoritarian systems, where parties often field a single candidate per district, elections tend to favor candidates with extensive experience, financial resources, and political networks—typically older men (Lee and McClean 2022) from the dominant ethnic group (Henig and Henig 2001). By contrast, under proportional rules, where parties present lists of candidates, parties have incentives to nominate a more diverse slate of candidates that can maximize their collective appeal (Norris 2004). Even within mixed-member systems, the reform from a more majoritarian to a more proportional system can play out in favor of politically underrepresented groups because smaller parties have a greater opportunity to win seats from the PR-tier. In many democracies, parties tend to adopt quotas for candidates from underrep-

resented groups specifically for the PR-tier, such as quotas for women or youth (Belschner 2021; Clayton et al. 2017). Women and young people may therefore advocate for proportional rules for the instrumental purpose of increasing their descriptive representation—that is, the extent to which the legislative body resembles these groups—which in turn can increase their political efficacy (Hayes and Hibbing 2017; West 2016) and the degree to which policy outcomes reflect their interests (Clayton et al. 2017; McClean 2025; Schwindt-Bayer 2006).

However, this is not to suggest that young and female voters do not care about accountability. To the contrary, these groups may have a heightened awareness of the ways in which those in power can abuse their authority, whether through neglect, bias, or outright corruption. Indeed, several studies find that women and young people are less tolerant than others of corruption and more willing to punish corrupt officials (Barnes and Beaulieu 2019; Eggers et al. 2018; Esarey and Schwindt-Bayer 2017; Hughes 2011; McClean and Ono 2022; Sloam 2014). Moreover, these voter groups tend to punish corrupt politicians more when the latter are from their group because they are more engaged when assessing these politicians (Jones 2014) and strongly dislike it when politicians from their group are portrayed as less honest (Eggers et al. 2018). Members of these communities are also particularly vulnerable because while they are often more likely to experience the negative impact of policy decisions, they are typically less likely to have the necessary resources or political power to hold politicians accountable for their actions.

With that being said, we still expect that young and female voters will prioritize increasing representation over accountability when it comes to electoral reform. An important, yet under-emphasized dimension to the debate over electoral systems and their consequences is that while majoritarian systems may increase accountability, this does not mean that representatives will be *equally* accountable to all voters—particularly those that have historically lacked representation. Without a seat at the table, underrepresented groups may often be ignored in the political process, which can limit the effectiveness of accountability mechanisms

in protecting their interests. By first increasing representation through the introduction of more proportional electoral rules, these groups can ensure that their interests are incorporated from the outset of the policymaking process, making accountability mechanisms more likely to be effective.

> *H1. Younger and female voters will prefer more proportional rules than older and male voters.*

Although we anticipate that underrepresented voters will generally favor more proportional rules, there are several reasons why we expect to observe variations in support within these groups.[2] First, while we anticipate that young and female voters will be more likely to support rules emphasizing representation over accountability, we expect these effects will be stronger when it comes to more major electoral rules, such as the overall choice of electoral system, than for more detailed, minor rules. This is because electoral reform tends to be an elite-driven process, shaped by inter-party bargaining (Laycock et al. 2013), where parties negotiate the specific details of reform. Voters, by contast, tend to focus more on the general electoral rule—such as whether it is an MMM or MMP system—and care less than parties do about these technical details. As Renwick (2017, 346) puts it, from the practitioner's point of view, academic research is simply "prone to overestimating the degree of engagement with politics and political choices that most voters want." It is also possible that this discrepancy in support stems from differences in knowledge about the technical aspects of electoral rules across these groups.

Moreover, details of the new electoral system may still be open to further discussion even at the elite level and can be decided after the election, as major parties often seek voter support for these details through the electoral process. For instance, when South Korea's two major parties agreed to pass an electoral reform bill a year before the 2020 general election, they reached a consensus that the bill should strengthen the proportionality and

[2]We formulated these additional expectations concerning heterogeneous effects after accessing the data, so they are not included in our original pre-registration plan. See Appendix for more detail.

representativeness of the country's voting rules—specifically by combining MMM and MMP rules, with the former applying to 36% and the latter to 64% of the PR-tier seats. Although the 2020 general election was scheduled to be held under this new system, both the ruling and major opposition parties agreed to re-discuss the details of the new rules—such as the possible expansion of the National Assembly by increasing the number of PR-tier seats—after the 2020 general election, as they viewed them as important yet politically sensitive issues.

In sum, since voters tend to pay greater attention to more major electoral rules and are typically less interested in and less knowledgeable about minor rules than elites, the strength of young and female voters' preferences for these different types of proportional rules is likely to differ.

> *H2. The preference for more proportional rules among young and female voters will be stronger regarding more major electoral rules than for more minor rules.*

Next, the literature on voting behavior suggests that partisan cues significantly influence voters' choices regarding institutional rules (e.g., Laycock 2013; Lupia and McCubbins 1998). However, for women and young people—who are often sidelined in political arenas shaped by majoritarian systems and dominated by major parties—party identification may not fully encapsulate their attitudes toward electoral systems and the inherent trade-offs between representation and accountability. We expect that these groups' evaluations of electoral rules will be guided not solely by their partisan leanings but also by their perceived political standing, with gender and age continuing to shape their preferences even after accounting for partisan differences.

Furthermore, this pattern may be even more pronounced among women and young people who support minor parties or identify as independents. Facing a double layer of underrepresentation—stemming from both their partisan affiliation and demographic attributes—these groups may feel an even stronger motivation to support more proportional rules that

promise greater political inclusion. Consequently, we predict that among minor party supporters and independents, this compounded marginalization will make young and female voters' preference for representation distinctly more pronounced than among those who support major parties.

> *H3. Young and female voters' preference for more proportional rules will be more evident among independent or minor party supporters than among major party supporters.*

Context: 2020 Electoral Reform in South Korea

Electoral institutions do not always produce context-invariant effects but often interact with social cleavages, voter preferences, and electoral geography (e.g., Amorim Neto and Cox 1997). For this reason, it is important to situate our study within the South Korean context.

While young and female voters are politically underrepresented in most countries, they face a particularly challenging environment in South Korea. In terms of gender equality, South Korea lags behind other developed countries. For instance, among the 38 member nations of the Organization for Economic Cooperation and Development (OECD), South Korea is ranked 29th according to The Economist's glass-ceiling index, which measures the role and influence of women in the workforce.[3] Korea also has the largest gender wage gap, lowest share of women on the boards of publicly listed companies, and fifth-lowest percentage of female legislators (19%) in the OECD.[4] In addition, within the central government, less than 25% of bureaucrats in managerial positions (grade 1-4) are women.[5]

As for age equality, young Koreans also have relatively low levels of political participation and representation compared to other age groups. In the 2020 general elections, less than

[3] "The Economist's Glass Ceiling Index," The Economist, https://www.economist.com/graphic-detail/glass-ceiling-index (accessed January 31, 2025).

[4] Don S. Lee and Charles T. McClean, "South Korea's new leader says there's no gender inequality problem," The Washington Post, April 26, 2022.

[5] 24.4% of bureaucrats at grade 4 are women, but in higher ranks, only 10% are women. See Ministry of Personnel Management, https://www.mpm.go.kr/mpm/comm/newsPress/newsPressRelease/ (accessed accessed January 31, 2025).

50% of adults in their 20s and 30s voted compared to over 70% of those age 60 or older.[6] Young adults are also greatly underrepresented in the national legislature. Just 17 (5.7%) of the 300 members are age 40 or under, well below the OECD average of 23%,[7] and the average age of legislators from the two largest political parties is 55.4.[8] This age gap in politics has led to an active discussion about the need for greater social protection programs for the young, such as a youth basic income, to enhance their civic engagement. For this reason, public opinion polls conducted before the adoption of electoral reform indicate that voters in the 20s (41%) and 30s (53%) show stronger support for more proportional rules versus the current majoritarian rules than those in their 50s (31%) and 60s or older (24%).[9]

In the initial years following South Korea's democratization in 1988, citizens elected their national lawmakers via an MMM system, also known as parallel voting, where most representatives were elected by first-past-the-post voting (FPTP) and a few were elected via party-list proportional representation (PR).[10] After several peaceful power transfers between liberal and conservative parties, however, the demand for reform of the electoral system to one that strengthens proportionality and representation began to grow among civil society and minor opposition parties. In 2015, a report from the National Election Commission of Korea suggested the more representative MMP system, used in Germany, as a model for electoral reform.[11] However, the negotiation process among legislative parties was not

[6] "Voter Turnout in the 15th General Election... Has the Decisive Power Increased?" KBS News, https://news.kbs.co.kr/news/view.do?ncd=3268098 (accessed January 31, 2025).

[7] "Republic of Korea National Assembly," Inter-Parliamentary Union Parline Database, https://data.ipu.org/parliament/KR/KR-LC01/; "Youth Empowerment & Intergenerational Justice," The OECD, https://www.oecd.org/en/topics/sub-issues/youth-empowerment-and-intergenerational-justice.html (accessed January 31, 2025).

[8] In Korean politics, the two main parties tend to win the vast majority of legislative seats. For example, in 2020, the ruling Democratic party and opposition United Future party captured 283 (94.3%) of 300 seats between them.

[9] Gallup Korea Daily Opinion, https://panel.gallup.co.kr/Contents/GallupKoreaDaily/GallupKoreaDailyOpinion_355(20190524).pdf (accessed January 31, 2025).

[10] For example, in the 2016 legislative election, 84.3% (253 members) was elected by FPTP voting, while 15.7% (47 members) was elected by PR voting.

[11] Compared to the Korean MMM system, the German MMP system was viewed as a more proportional one, because in the latter, the number of seats is compensated for a small party in proportion to the number of votes received by the small party in the PR-tier, in consideration of the unequal distribution of seats in constituencies typically favoring large parties and unfavorable to small parties. In the former, however, there is no such thing as compensatory-proportional allocation. See Shugart and Wattenberg (2001, 14) for

easy or smooth. While small parties such as the Justice Party were strongly supportive of reform, the two major parties were initially skeptical (ruling Democratic Party) or stridently opposed (opposition Mirae Party). After years of fierce disputes between the major parties, the commission's report eventually received the backing of the ruling Democratic Party, the largest party in the National Assembly. Over time, even the main conservative Mirae Party became somewhat more open to the reform, but continued to express reservations. Ultimately, the electoral system reform bill was passed on December 23, 2019. Throughout this process, however, the ruling and opposition parties continued to discuss key details of the new electoral system, including whether to expand the size of the National Assembly and how to decide the ratio of seats between constituencies and the PR tier.

Public opinion surveys show that Korean citizens overall seem to support a move toward a more proportional voting rule. According to a survey conducted in December 2018, 66.3% of respondents were in favor of reforming the electoral system to increase the representativeness of the people, while 73.4% answered "the status quo, more majoritarian system does not reflect the will of the people well."[12] The results from another survey conducted in November 2018 indicate that 58.2% of respondents support the reform of the electoral system into a more proportional one as they want rules for electing members of the National Assembly to minimize wasted votes and distribute seats proportional to votes received.[13] When breaking down these polls into subgroups, positive responses still prevailed across regions, ideological orientations, and party identification.[14] However, a majority of voters disapproved of expanding the size of the National Assembly, which might be necessary in the process of reforming the electoral system, even if they supported the adoption of a more proportional voting rule via electoral reform (53%) or even if it were conditional on reducing the budgets

details.

[12] "66% In Favor of 'Election Reform'... 79% 'Opposes Expanding the Number of Members"', KBS News, https://news.kbs.co.kr/news/view.do?ncd=4107659 (accessed January 31, 2025).

[13] "58% In Favor of Electoral System Reform... 60% Opposes Expansion of the Number of Members," Seoul Economic Daily, https://www.sedaily.com/NewsView/1S73XQUWQ9 (accessed January 31, 2025).

[14] Ibid.

and privileges for members of the National Assembly (59.9%).[15]

Our conjoint experiment was conducted in the context of this electoral reform. Given that voters' preferences for electoral reform are multidimensional rather than unidimensional, choosing a conjoint experiment—rather than separately asking about support for electoral rules that feature each attribute—has both theoretical and practical advantages.

Experimental Design

We conducted our conjoint survey experiment with a nationally representative sample of more than 1,800 South Korean adults who were eligible voters from late March to early April 2020, about three months after the passing of the electoral reform bill and just before the legislative election on April 15, 2020. Our survey timing—immediately before an actual election with new rules—should help respondents to be primed to properly assess both the previous and new electoral systems, and to perceive our survey experiment as realistic and take it seriously. The survey sample was recruited by Research & Research, one of the main survey research firms in South Korea. The survey design and administration were conducted via Qualtrics, and Research & Research ensured the recruitment of a nationally representative sample and the overall implementation of our survey.[16]

Conjoint survey experiments have been increasingly and actively used among electoral scholars. One of the greatest advantages of conjoint analysis is the ability to identify "which components of the manipulation produce the observed effects" (Hainmueller, Hopkins, and Yamamoto 2014, 2). Voters' choices and election outcomes are the results of aggregation. That is, voters choose candidates and parties based on a range of attributes and policy positions. However, since we are only able to observe which candidates and parties receive how many votes, we do not know why an individual voter chose a particular candidate or party—a fundamental challenge in understanding the multidimensional preferences for candidate

[15]Ibid.

[16]We pre-registered our research design, sampling, and analysis plan at OSF. The human research subjects aspect of our experimental protocol was approved by our university's Institutional Review Board.

Table 1: Conjoint Survey Design

Attribute	Level
Major Rule	
Electoral System	MMP
	Mix of MMM and MMP
	MMM*
Minor Rules	
Size of the National Assembly	Same Size, More Proportional
	Bigger Size, More Proportional
	Status Quo*
Aggregation of PR Votes	PR Votes Pool at National Level*
	PR and SMD Votes Pool at National Level
	PR Votes Pool at Provincial Level
Threshold to Obtain a PR Seat	1 Percent of PR Vote
	3 Percent of PR Vote*
	5 Percent of PR Vote
Dual Listing	No Dual Listing*
	Dual Listing

Note: MMP: Mixed Member Proportional; MMM: Mixed Member Majoritarian. For each attribute, levels are organized such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom, with a star (*) indicating the status quo level.

attributes and party policies behind vote selection in representative democracies (Horiuchi, Smith, and Yamamoto 2018, 191). With conjoint analysis, however, we can decompose treatment effects (i.e., candidate attributes and party policies), identify which elements of a multidimensional treatment exert influence on voters' decisions (Hainmueller, Hopkins, and Yamamoto 2014), and determine how subgroups respond to these elements differently (Leeper, Hobolt, and Tilley 2020). In the case of electoral reform, conjoint analysis helps us understand the fundamental reason behind why voters chose one electoral system over the other by identifying which specific components of a multidimensional electoral system citizens prefer—something that is not possible with traditional unidimensional survey experiments.

We designed our conjoint experiment both to test our hypotheses and to reflect the actual debate surrounding the 2020 electoral reform in Korean politics. At the heart of this debate was the balance between accountability and representation, a theme that political elites and media outlets repeatedly invoked. To capture this debate empirically, we focused on five attributes that were central to the reform (see Table 1).[17] For each attribute, we selected levels that represented key points of contention. Most attributes have three levels, with the exception of Dual Listing (two levels). In the survey, we randomized the order of attributes and values within attributes at the respondent level.

Because our goal was to evaluate how underrepresented groups view the accountability-representation trade-off, it was important that respondents approached the conjoint task with a shared interpretive frame. Although our survey was conducted immediately before the 2020 general election—when the electoral reform was highly salient and many participants were already familiar with the issues—we sought to encourage consistency by priming respondents in the question wording to consider this trade-off directly. We then reinforced this frame by providing an information table that explained each rule and located it along the accountability-representation spectrum, helping to minimize knowledge gaps and provide respondents with a shared point of reference (see Appendix Figures A1 and A2).

Of the five attributes listed in Table 1, the first concerns the major rule under debate: the overall choice of electoral system. Along the accountability and representation spectrum, South Korea's previous MMM system, which was used until the 2016 general election, placed the greatest emphasis on accountability, while the MMP system that was being considered during the reform process is the one most favorable toward representation. The electoral system newly adopted in 2020—a mix of MMM and MMP, where the compensatory-proportional allocation of the MMP system only applies to 64% of the PR-tier seats—is positioned between the two mixed-member systems, as it represents a balance between accountability and

[17]Table 1 distinguishes between major and minor rules for the purpose of clarifying how we test H2, but these labels were not provided to respondents in the conjoint experiment itself. We also asked respondents about their preferences regarding institutional reforms of Korea's executive branch but focus only on electoral reforms for the legislative branch in the main text (see Table A15).

representation.

The other four attributes in our conjoint represent the more minor electoral rules up for debate. The second attribute is related to the size of the national legislature. If the number of seats was increased in the National Assembly by expanding only the PR-tier, then this reform would increase representation without large costs for accountability by keeping the number of SMD-tier seats the same. However, if the National Assembly was kept at the same size by reducing the number of SMD-tier seats by an amount equal to the increase in PR-tier seats, then the increase in representation would be accompanied by a decrease in accountability. As such, we have placed the latter reform as closer to the representation end of the spectrum with the former reform positioned in the middle.

The third attribute concerns the aggregation of PR votes. When PR votes are aggregated at the national level rather than at smaller provincial levels, parties compete for a larger number of seats. This lowers the effective threshold and makes it easier for smaller parties to win representation in the PR tier. By contrast, aggregating PR votes within provinces tends to generate more "overhang seats"—constituency seats a party wins in excess of those implied by its PR vote share—because compensation is calculated within a smaller pool. Since overhang seats tend to go to larger parties, provincial-level aggregation shifts the system more toward accountability, whereas national aggregation yields fewer overhang seats and thus greater representation (Reynolds et al. 2008).

The fourth attribute is the threshold to obtain a seat via PR-tier votes. Here, the logic is straightforward: the higher the threshold, the greater accountability the system has, because it becomes more advantageous to larger parties. Conversely, the lower the threshold, the more representation the system has, because it opens the door to smaller parties.

The last attribute is dual listing, which allows candidates to be listed in both SMD- and PR-tiers. While this system is likely to help both larger and smaller parties, which can also list popular constituency candidates in the PR-tier, increasing the visibility of their list of PR candidates, larger parties may have greater advantages given their sizable pool of popular

constituency candidates.[18] Therefore, approving dual listing will increase accountability, but disapproving it will be better for representation.

Our specific question wording, information table, and conjoint experiment are shown in Appendix Figure A1 and Table A3 (translated from Korean to English).[19] Respondents were asked to answer up to five times regarding which system profile they would prefer, with a new set of attribute values for each profile being presented through randomization each time. In total, more than 2,500 survey experiment forms were handed out by Research & Research, and among them, 1,845 were completed in full.

In addition to the question measuring individual voters' preferences regarding electoral reform and institutional rules, we included questions to obtain respondents' demographic, socioeconomic, and political characteristics, such as gender, age, income level, and party identification. In Appendix Tables A1 and A2, we report descriptive statistics regarding the characteristics of survey respondents by gender and age.

Results

Do young and female voters prefer rules that prioritize representation over accountability? We have argued that these voter groups will do so (H1) for several reasons discussed above, which would help to place them in a better position than the status quo. We further discussed our predictions that we might observe variations in support within these groups: 1) the preference for more proportional rules among young and female voters will be stronger regarding the overall choice of the electoral system than for more minor rules (H2); and 2) young and female voters' support for more proportional rules will be more evident among independent or minor party supporters than among major party supporters (H3).

Overall, we find support for most of our predictions, with some notable exceptions. In

[18] An alternative view is that dual listing might do more to benefit smaller parties, which often struggle to recruit strong constituency candidates and instead place their most competitive candidates on the PR list. However, in designing the survey, we emphasized the potential advantages for larger parties and for promoting accountability. As such, we expect this framing reflects how respondents were likely to interpret this attribute in the conjoint task.

[19] Appendix Figure A2 and Table A4 show the Korean-language version of our experiment.

the following sections, we undertake a thorough analysis of these patterns, beginning with a collective overview of participant responses, before delving into the nuanced distinctions influenced by respondent gender, age, and political affiliation.[20]

General Patterns

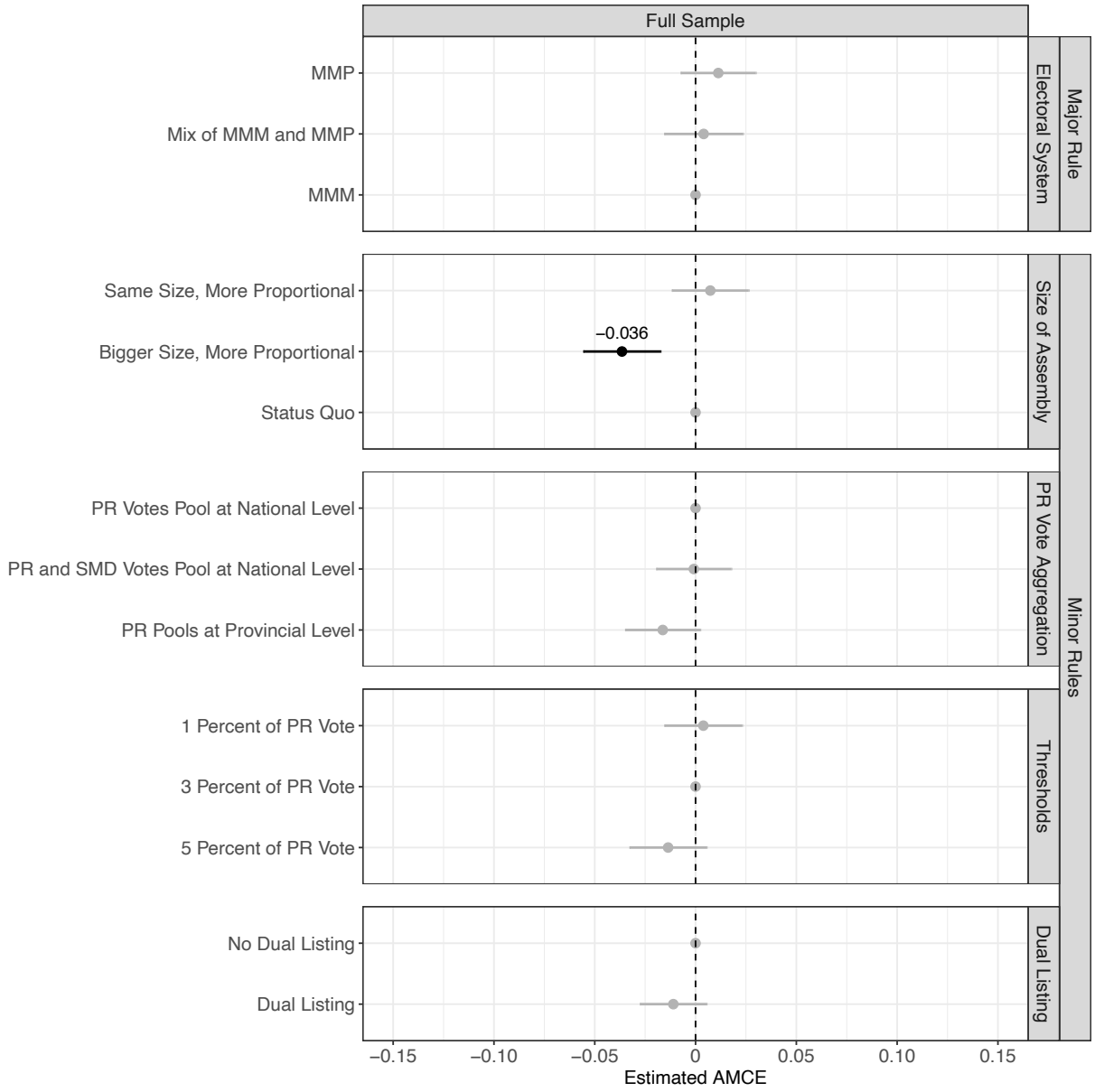
Figure 1 shows the estimated average marginal component effects (AMCEs) for the full sample of respondents.[21] The circles represent the point estimates, and the bars show 95% confidence intervals with standard errors clustered at the respondent level. For each attribute of our hypothetical electoral system profiles, we set the baseline to be the status quo and organize the levels (i.e., specific system rules) so that they range from the level that most emphasizes representation at the top to the one that most emphasizes accountability at the bottom. Positive AMCEs indicate that respondents prefer the respective rule compared to the status quo baseline, while negative AMCEs suggest respondents prefer the baseline. Significant estimates ($p<0.05$) are highlighted in black, while insignificant ones are grey.

We find no significant support among respondents for adopting any specific electoral rules (major or minor) that would shift Korea's system away from the status quo. However, there is significant opposition to implementing more representative rules in one attribute category: the average participant in our survey disliked the idea of simultaneously increasing the size and proportionality of the assembly. Relative to Status Quo for Size of Assembly, respondents were 3.6 percentage points less likely to support Bigger Size, More Proportional.

[20] In the Appendix, we test two additional pre-registered hypotheses: that support for more representative electoral rules will be stronger among voters of lower socioeconomic status and supporters of President Moon Jae-in, who was favorable toward Korea's proposed electoral reforms. We find modest support for the first expectation, with lower income voters marginally more likely to support a more proportional National Assembly ($p<0.10$), and stronger support for the second, as Moon supporters are significantly more likely to prefer more representative electoral rules and a more proportional assembly (Tables A13 and A14).

[21] Recent work has raised questions about how AMCEs should be interpreted, noting that it may not directly reflect majority preferences (Abramson et al. 2022). Others argue, however, that the AMCE provides a meaningful aggregation of both the direction and intensity of multidimensional preferences (Bansak et al. 2023). We follow this latter view, using the AMCE to summarize how voters evaluate specific institutional attributes, while recognizing that it does not directly capture majority will.

Figure 1: General Patterns in Electoral System Preferences



Note: Average marginal component effect (AMCE) estimates with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A5.

Underrepresented Voters

In this section, we investigate our results by gender (Figure 2), age (Figure 3), and the intersection of gender and age (Figure 4). In Figure 2, the left panel shows the AMCEs for

female respondents, the center panel shows the AMCEs for male respondents, and the right panel shows the estimated differences between women and men.[22] Figure 3 follows a similar presentational style, dividing respondents under 40 (left panel) from those 40 or older (center panel) using the age cutoffs commonly found in OECD reports on youth empowerment.[23] The right panel again reflects the estimated differences across subgroups. Finally, Figure 4 adopts the same three-panel format to examine the intersection of gender and age, comparing our theoretically most underrepresented group—young women under 40—to the least underrepresented group—older men 40 and above. A positive estimate in the right panels thus indicates that women, younger adults, or young women tend to prefer the given electoral rule (over the status quo) compared to men, older adults, or older men, respectively.[24]

Overall, based on the patterns observed in Figures 2, 3, and 4, we find some support for H1 and H2.[25] First, women and young adults tend to demonstrate a stronger preference for more proportional rules compared to male and older respondents (H1). Furthermore, the preference for more proportional rules among female and young voters is stronger when it comes to the major electoral rule—the overall choice of the electoral system—than for more minor rules, such as PR Vote Aggregation and Dual Listing (H2). The largest magnitude effects appear for the intersection of gender and age, where young women under 40 show the strongest preference for more proportional rules. Specific results are discussed below.

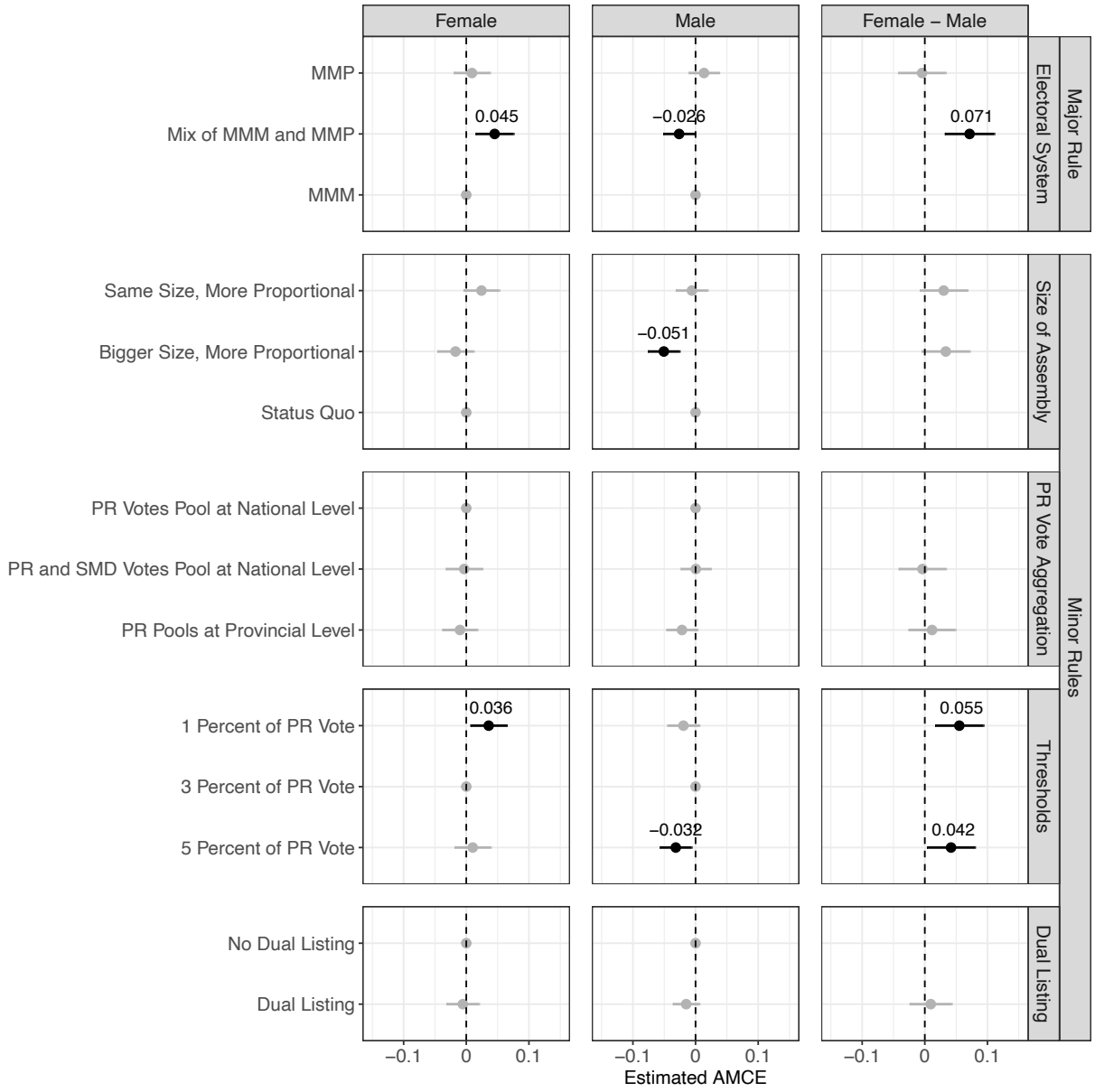
[22]Leeper, Hobolt, and Tilley (2020) suggest that researchers should use marginal means rather than AMCEs for estimating differences between subgroups, especially since the choice of baseline in many studies is arbitrary. In our case, we choose to focus on AMCEs because the baseline choice is not arbitrary: We are theoretically interested in preferences relative to the status quo electoral rules. We find similar results if we instead use marginal means.

[23]See https://www.oecd.org/en/topics/sub-issues/youth-empowerment-and-intergenerational-justice.html (accessed January 31, 2025). Public opinion surveys and academic research on youth also frequently classify individuals under 40 as part of the 'youth' category for analytical purposes, particularly when discussing youth representation in national legislatures (e.g., McClean 2023).

[24]It is important to note that while our conjoint design enables us to account for multidimensional preferences among underrepresented voters, it does not allow us to make causal claims about the impact of gender or age on preferences, since we cannot randomly assign these characteristics to respondents.

[25]We also test the regularity assumptions commonly invoked in conjoint analyses and find no evidence of profile order effects, carryover effects, or fatigue effects (Bansak et al. 2018; Ham, Imai, and Janson 2023; Hainmueller, Hopkins, and Yamamoto 2014), which gives us further confidence in our results.

Figure 2: Electoral System Preferences by Gender



Note: Average marginal component effect (AMCE) estimates with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A6.

Gender

Beginning with Figure 2, we find evidence that women tend to be more favorable than men toward institutional reforms that emphasize representation over accountability, while men

are more likely to prefer the status quo. However, the strength of this relationship varies substantially across different electoral system attributes, suggesting that women may be hesistant to fully transition away from rules that promote accountability.

Turning to the left panel, we find that female respondents prefer more representative electoral rules in two attribute categories: Electoral System and Thresholds. For Electoral System, women were 4.5 percentage points more likely to support the more representative Mix of MMM and MMP compared to the status quo MMM system, which emphasizes accountability. However, women did not support a full shift to the most representative MMP system. For Thresholds, women were 3.6 percentage points more likely to favor reducing the threshold from 3 Percent to 1 Percent of PR Vote, which would boost representation. There is also suggestive evidence that women prefer a more representative rule for Size of Assembly, as female respondents were 2.4 percentage points more likely to select electoral system profiles with Same Size, More Proportional ($p<0.1$).

In contrast, male respondents did not support electoral rules that would increase representation at the cost of accountability in any of the five categories (center panel). Moreover, in two of the categories where women preferred implementing more representative rules—Electoral System and Size of Assembly—men significantly opposed such reforms. The clearest gender difference can be seen regarding the Electoral System attribute. In direct contrast to the preferences of women, men were 2.6 percentage points less likely to support a Mix of MMM and MMP compared to the current MMM system. Additionally, while women indicated a modest preference for increasing the proportionality of the assembly, men were 5.1 percentage points less likely to favor Bigger Size, More Proportional.[26]

The right panel of Figure 2 shows that a number of these described gender differences are statistically significant. In terms of prioritizing representation over accountability, women

[26]While men in our survey opposed instituting more representative rules, in most categories they indicated a preference for preserving the status quo over implementing reforms that would instead increase accountability. For example, men were 3.2 percentage points less likely to support raising the threshold from 3 Percent to 5 Percent of the PR Vote and 2.2 percentage points less likely to back changing PR Vote Aggregation to PR Pools at Provincial Level, although the latter result is only marginally significant ($p<0.10$).

were 7.1 percentage points more likely to back a Mix of MMM and MMP, and 5.5 percentage points more likely to prefer a 1 Percent of PR Vote threshold. There is also suggestive evidence that women are more favorable toward Bigger Size, More Proportional ($p<0.1$), while the coefficient for Same Size, More Proportional is positive but outside conventional levels of significance ($p=0.12$). The one exception to this pattern is that women were 4.2 percentage points more likely than men to favor 5 Percent of PR Vote, which would increase accountability. However, this result stems from men's stronger preference for the status quo (center panel) rather than women's preference for increasing accountability (left panel).[27]

Age

Turning to Figure 3, we find that young people are more supportive of institutional features that emphasize representation, but only regarding the choice of Electoral System. Compared to the status quo MMM system, adults under 40 were 4.6 percentage points more likely to prefer fully adopting an MMP system, the most representative of the three choices.[28] As for the other four attributes, young people in our survey demonstrated no preference for more representative rules.

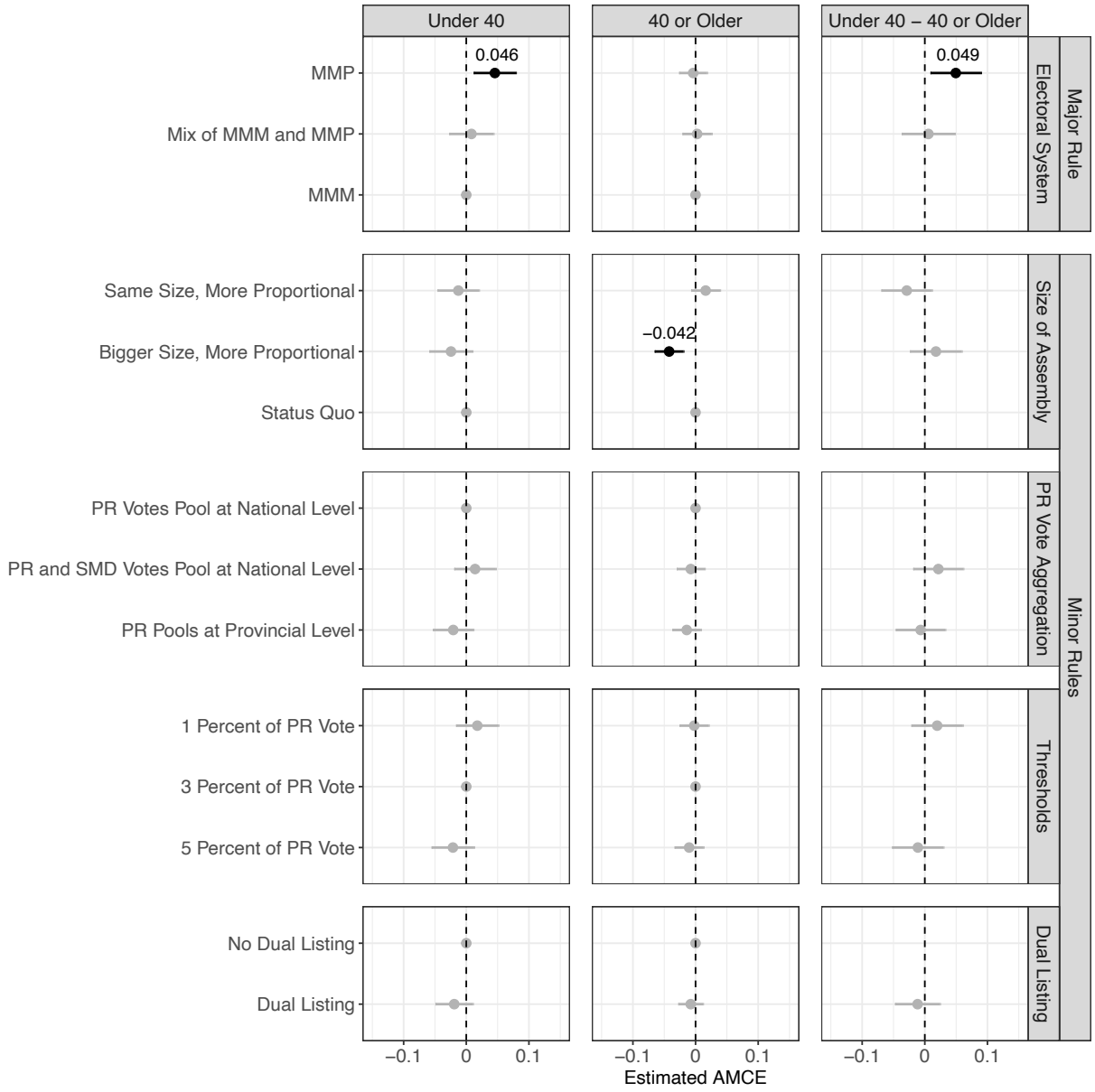
There are also few significant differences among older voters (center panel). Similar to our findings for male respondents, respondents 40 or older appear to prefer the status quo over reforms in either direction. They showed no support for adopting more representative electoral rules at the cost of accountability, particularly when it came to Size of Assembly, where they were 4.2 percentage points less likely to support Bigger Size, More Proportional.

The right panel of Figure 3 confirms that the age difference regarding the overall choice of Electoral System is statistically significant. Respondents under 40 were 4.9 percentage

[27]In addition to testing for pairwise differences in AMCEs between men and women, we also conduct nested model comparison tests, which use F-tests to evaluate whether models that account for subgroup differences better fit the data than models with only the conjoint attributes (Leeper, Hobolt, and Tilley 2020). In line with the pairwise tests, we find that women and men differ significantly in their preferences toward Electoral System ($F = 6.59$, $p<0.10$) and Thresholds ($F = 2.93$, $p<0.05$).

[28]In a nested model comparison test, we similarly find that only Electoral System ($F = 2.60$, $p<0.10$) reaches even marginal levels of significance, further suggesting that the only substantial difference across age groups is young adults' greater support for MMP over the current MMM system

Figure 3: Electoral System Preferences by Age

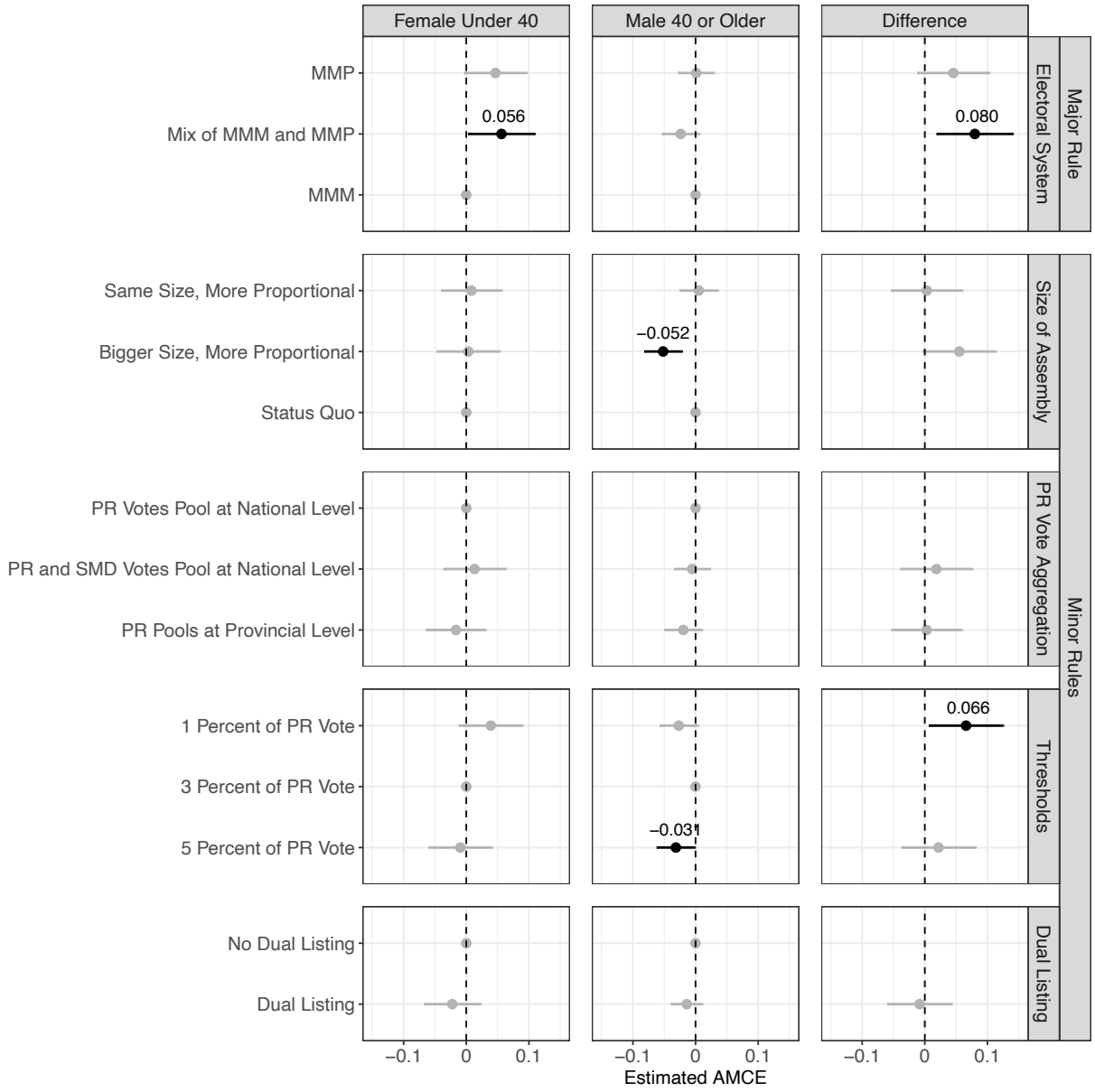


Note: Average marginal component effect (AMCE) estimates with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A7.

points more likely than respondents 40 or older to support shifting Korea toward the most representative MMP system.[29]

[29]If we disaggregate the results by age decade (Table A8), we find the strongest support for MMP is among those aged 30-39, while the weakest support for revising major rules is among those 50 or older, who also

Figure 4: Electoral System Preferences by Gender and Age



Note: Average marginal component effect (AMCE) estimates with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A9.

show a strong dislike for increasing the size and proportionality of the assembly.

Gender and Age

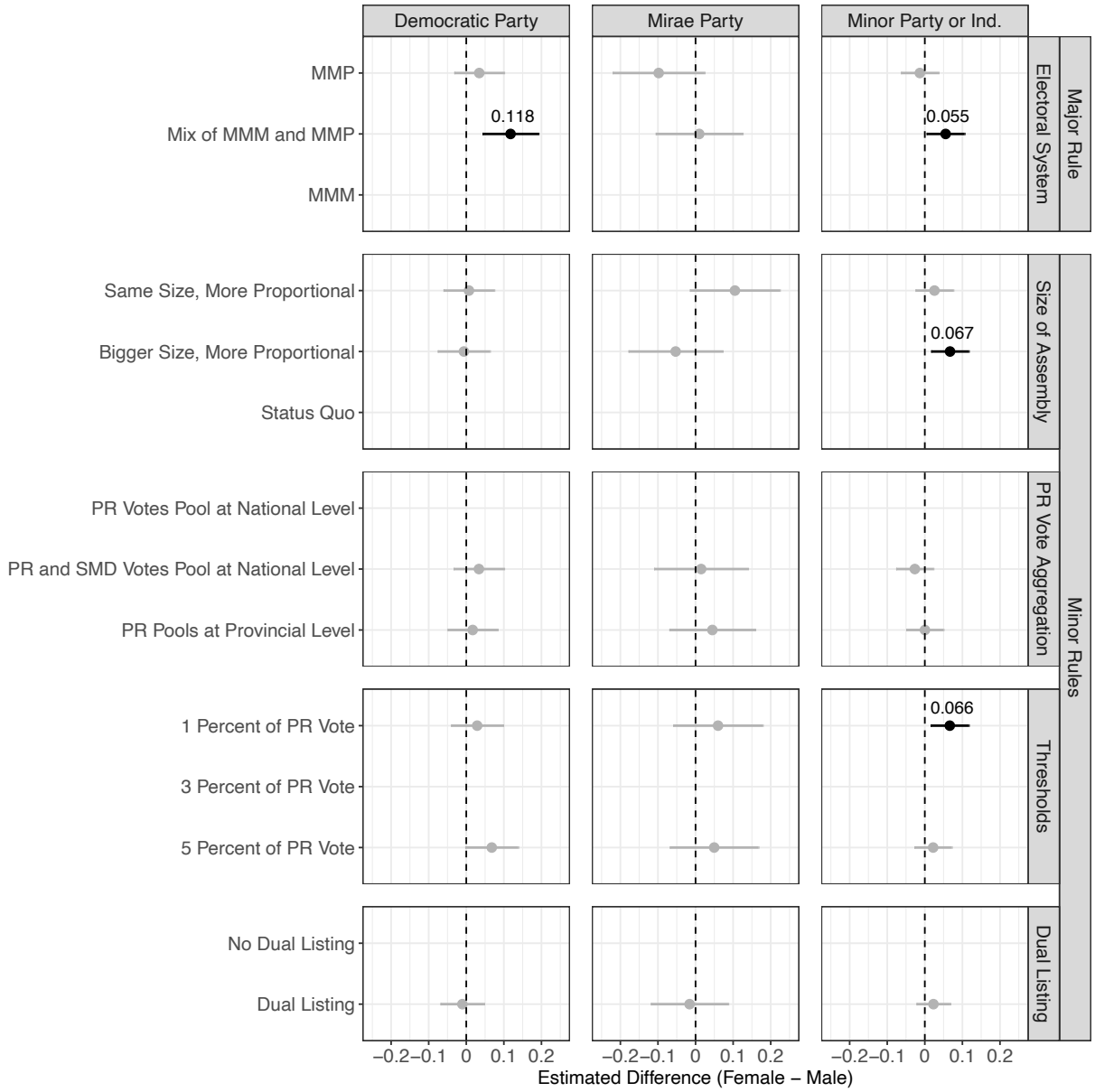
Finally, we consider the intersection of gender and age (Figure 4). We find that young women are significantly more supportive of proportional rules compared to older men. Women under 40 were 5.6 percentage points more likely to favor a mix of MMM and MMP compared to the status quo, and there is suggestive evidence that they are 4.7 percentage points more likely to choose MMP over the status quo ($p<0.10$). By contrast, men age 40 or older were 5.2 percentage points less likely to support increasing the size and proportionality of the assembly, though they were also 3.1 percentage points less likely to favor increasing the threshold to 5 Percent of PR Vote. Several differences are again statistically significant, with young women being 8.0 percentage points more likely to support a Mix of MMM and MMP and 6.6 percentage points more likely to favor reducing the threshold to 1 Percent of PR Vote. Notably, we find the largest differences in support for representation over accountability when comparing the most underrepresented group (younger women) with the least underrepresented group (older men).[30]

The Conditioning Effect of Partisanship

Is the greater support among women and young people for more representative rules evident even among members of the same party? As discussed above, we think that partisan perspectives may play a significant role in voters' assessments of electoral rules. However, given that young and female voters have historically lacked political voice and might not have been properly represented within the current majoritarian system, which is dominated by major parties, we anticipate that their assessments will be affected not only by their partisan attachments but also by their political status based on their demographic characteristics. If this is indeed the case, more evident differences by gender and age may exist within supporters of smaller parties and among independent voters (H3).

[30]In Table A9, we provide further evidence that older men seem the least supportive of more proportional rules among all four intersectional groups. For instance, older women were 4.0 percentage points more likely to favor a Mix of MMM and MMP, while younger men were 4.5 percentage points more likely to support a shift to MMP ($p<0.10$).

Figure 5: Gender Differences in Electoral System Preferences by Party



Note: Estimated differences in average marginal component effects (AMCEs) between women and men with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A11.

To test whether group differences are detectable within parties, Figures 5, 6, and 7 show our results for gender, age, and the intersection of gender and age by party affiliation. We divide respondents into three groups depending on their self-reported partisan support:

Democratic Party (the major liberal party, in power as of 2020), Mirae Party (the major conservative party, in opposition as of 2020), and Minor Party or Independent. For space reasons, we report only the estimated differences in AMCEs between women and men, between adults under 40 and those 40 or older, and between women under 40 and men 40 or older, rather than the estimated AMCEs for each demographic group.[31]

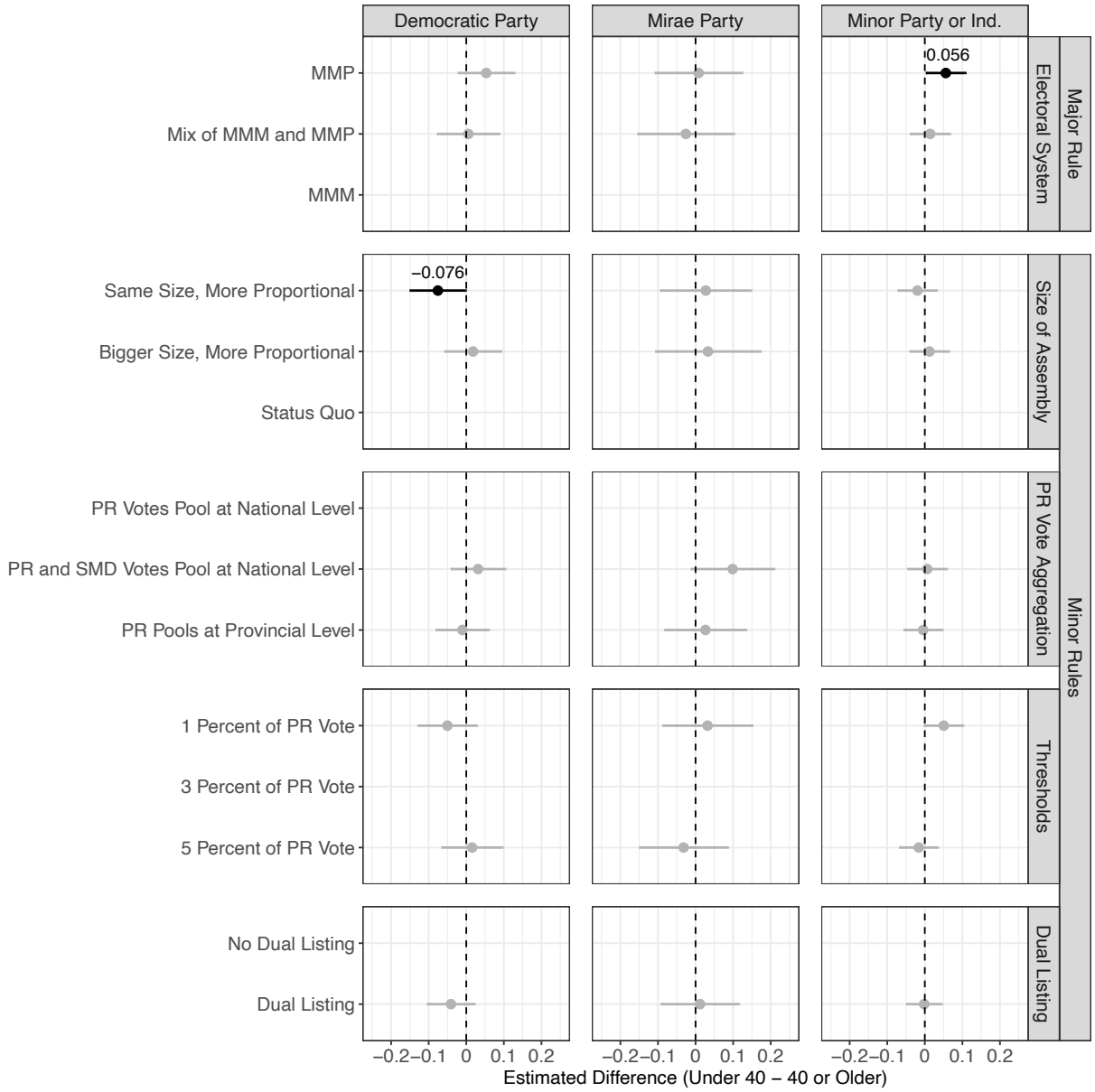
The overall results in Figures 5, 6, and 7 appear to support H3, as the clearest patterns concerning women and young people's greater emphasis on representation can be seen among minor party supporters and independents. While the degree to which women and young adults prioritize representation over accountability is at times weakened by their party affiliation, in other cases, we find even stronger within-party age and gender differences in preferences for representation than we did in the full sample.

Gender

Beginning with gender (Figure 5), we find that women's stronger support for institutional rules that emphasize representation over accountability is most consistent and clearest among Minor Party or Independent respondents. Within this group, women were 5.5 percentage points more likely than men to pick a Mix of MMM and MMP, 6.7 percentage points more likely to choose Bigger Size, More Proportional, and 6.6 percentage points more likely to select 1 Percent of PR Vote. By contrast, we find no significant gender differences among supporters of the opposition Mirae Party, although there is suggestive evidence ($p<0.10$) that female backers of the Mirae Party were more supportive of Same Size, More Proportional. As a notable exception, female respondents who backed the ruling Democratic Party, the major liberal party in South Korea, were 11.8 percentage points more likely than their male counterparts to support a shift from the current MMM system to a Mix of MMM and MMP.

[31] In Appendix Table A10, we present overall electoral system preferences across party affiliation subgroups.

Figure 6: Age Differences in Electoral System Preferences by Party

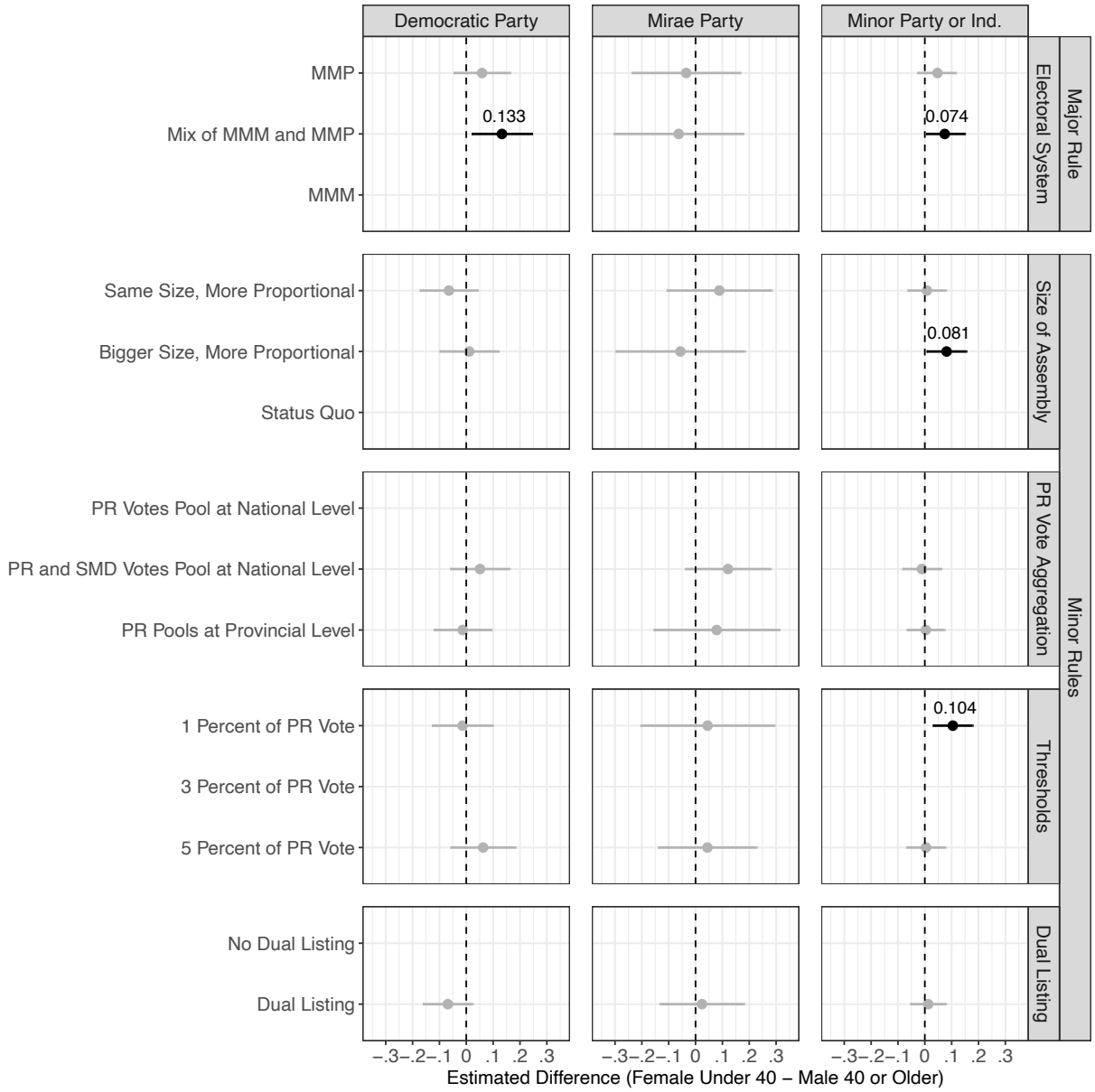


Note: Estimated differences in average marginal component effects (AMCEs) between respondents under 40 and those 40 or older with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A12.

Age

As for age (Figure 6), the evidence for young people's preference for institutional attributes promoting representation is statistically significant only among the Minor Party or Inde-

Figure 7: Gender and Age Differences in Electoral System Preferences by Party



Note: Estimated differences in average marginal component effects (AMCEs) between women under 40 and men 40 or older with clustered standard errors at the respondent level. Bars indicate 95% confidence intervals, while points without bars denote baseline categories. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. Significant estimates ($p<0.05$) are highlighted in black. For the table version of this figure, see Table A12.

pendent group. Adults under 40 in this group were 5.6 percentage points more likely than adults 40 or older to prefer an MMP system. By comparison, we again find no significant age differences among members of the Mirae Party, and an unexpected result for the Democratic

Party, where younger people were 7.6 percentage points less likely to favor Same Size, More Proportional for Size of Assembly.

Gender and Age

Finally, we again find the largest magnitude effects when examining the intersection of gender and age (Figure 7). Among Minor or Independent Party supporters, women under 40 preferred a Mix of MMM and MMP over MMM by 7.4 percentage points, Bigger Size, More Proportional over the Status Quo by 8.1 percentage points, and 1 Percent of PR Vote over 3 Percent of PR Vote by 10.4 percentage points. Similarly, we again find null results among Mirae Party supporters, while young women within the Democratic Party were 13.3 percentage points more likely than older men to favor a Mix of MMM and MMP over MMM.

In sum, while much recent experimental research considers the effects of electoral institutions on voter preferences to be shaped by individuals' partisan attachments, our findings suggest that partisan considerations alone may be insufficient—particularly for voter groups that have been consistently underrepresented in the current party system.

Discussion

Electoral reform often marks a pivotal moment for representative democracy. Yet, how such changes shape voter responses—particularly among those who are typically most affected by the reform—remains understudied. Leveraging a real-world electoral reform, where voters had to compare a more majoritarian (pre-reform) system with a more proportional (post-reform) one, our conjoint analysis sheds light on the preferences of politically underrepresented groups, specifically women and young people, regarding electoral rules and the trade-offs between representation and accountability. Overall, with some exceptions, we find that these groups tend to prefer more proportional rules that emphasize representation, with stronger effects when the choice concerns the overall electoral system rather than specific minor rules. After accounting for party identification, we find that within-party preferences for electoral rules that prioritize representation over accountability are clearer among women

and young people who are independent voters or support smaller parties than among those aligned with major parties.

Importantly, these groups did not simply cue on the reform option itself. Women and younger voters often favored alternatives that were even more representative than the rules adopted in 2020. This suggests their preferences reflected a broader orientation toward inclusiveness, rather than merely endorsing the elite-driven reform package.

At the same time, our design comes with certain limitations. To make sure respondents could evaluate complex institutional rules in a comparable way, we highlighted the accountability-representation trade-off in the prompt and provided an information table situating each rule along this spectrum. This strategy allowed us to focus directly on our theoretical question while also mirroring the public debate, but it also means our results reflect how voters responded when guided to think about this particular trade-off. Future research should examine whether respondents view electoral reforms in similar terms without such a prompt, as well as how accountability and representation are weighed against other considerations—such as legitimacy, expectations about policy outcomes, or respect for the rule of law—that may also shape preferences over institutional rules. Additionally, while we account for partisanship by examining age and gender differences within parties, our design cannot fully disentangle voters' underlying preferences for representation versus accountability from the influence of elite cues and partisan framing. Future research should investigate how these dynamics interact in shaping preferences for electoral reform.

Beyond partisanship, an additional question concerns differences within and across underrepresented groups themselves. In this article, we focused on women and younger voters as illustrative cases, but the mechanisms underlying their preferences may not be identical—for instance, age effects may partly reflect life-cycle dynamics or generational experiences, whereas gender gaps may stem from more persistent structural barriers to political inclusion. Disentangling these mechanisms and extending the analysis to other underrepresented identities, such as ethnicity or religion, would be a valuable direction for future research.

With these caveats in mind, our study offers both theoretical and empirical contributions to the literature by examining how two key dimensions of electoral institutions—representation and accountability—shape voter preferences, and by using a novel conjoint survey experiment within the context of a real-world electoral reform in a new but stable democracy in Asia, South Korea. However, one important qualification deserves attention: while women and young people preferred rules closer to the representation end of the spectrum, they did not want to abandon accountability altogether. This nuanced finding suggests that their preferences may reflect a search for an "electoral sweet spot" (Carey and Hix 2011): a moderate reform that enhances representation without imposing too heavy a cost on accountability.

Our experimental findings also complement recent observational work such as Becher and González (2019, 708), who identify another trade-off embedded in electoral rules: adopting more proportional systems can increase the "policy responsiveness of the legislature to the population at large," but at the potential cost of lowering the "quality of politicians." Although one might doubt whether voters take such institutional trade-offs into account when forming preferences about alternative electoral rules, our results provide experimental evidence that they do, at least when presented with clear information about those trade-offs.

Given that many young democracies in Asia and beyond often adopt winner-take-all systems for their legislative and executive elections within competitive party environments—and thus experience large gaps in the political representation of women and young people—our findings may provide valuable lessons for these countries as they consider reforms to make their electoral rules more proportional. As Lijphart (2012) argues, adopting more power-dispersing political institutions via electoral reform is crucial for democratic consolidation. At the same time, understanding how these reforms are perceived by voters is also essential to ensuring the effective functioning of representative democracy.

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How Do Underrepresented Voters View Electoral System Trade-Offs?

Don S. Lee and Charles T. McClean

Online Appendix

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1 Characteristics of Survey Respondents

Table A1: Characteristics of Survey Respondents by Gender

	Gender of Respondents					
	Women		Men			
	Mean	SD	Mean	SD	Difference	SE
Under 40	0.33	(0.47)	0.29	(0.45)	0.05*	(0.02)
Party Affiliation						
Democratic Party	0.31	(0.46)	0.30	(0.46)	0.01	(0.02)
Mirae Party	0.09	(0.28)	0.13	(0.33)	-0.04***	(0.02)
Minor Party or Independent	0.60	(0.49)	0.57	(0.49)	0.03	(0.02)
N	699		967			

Table A2: Characteristics of Survey Respondents by Age

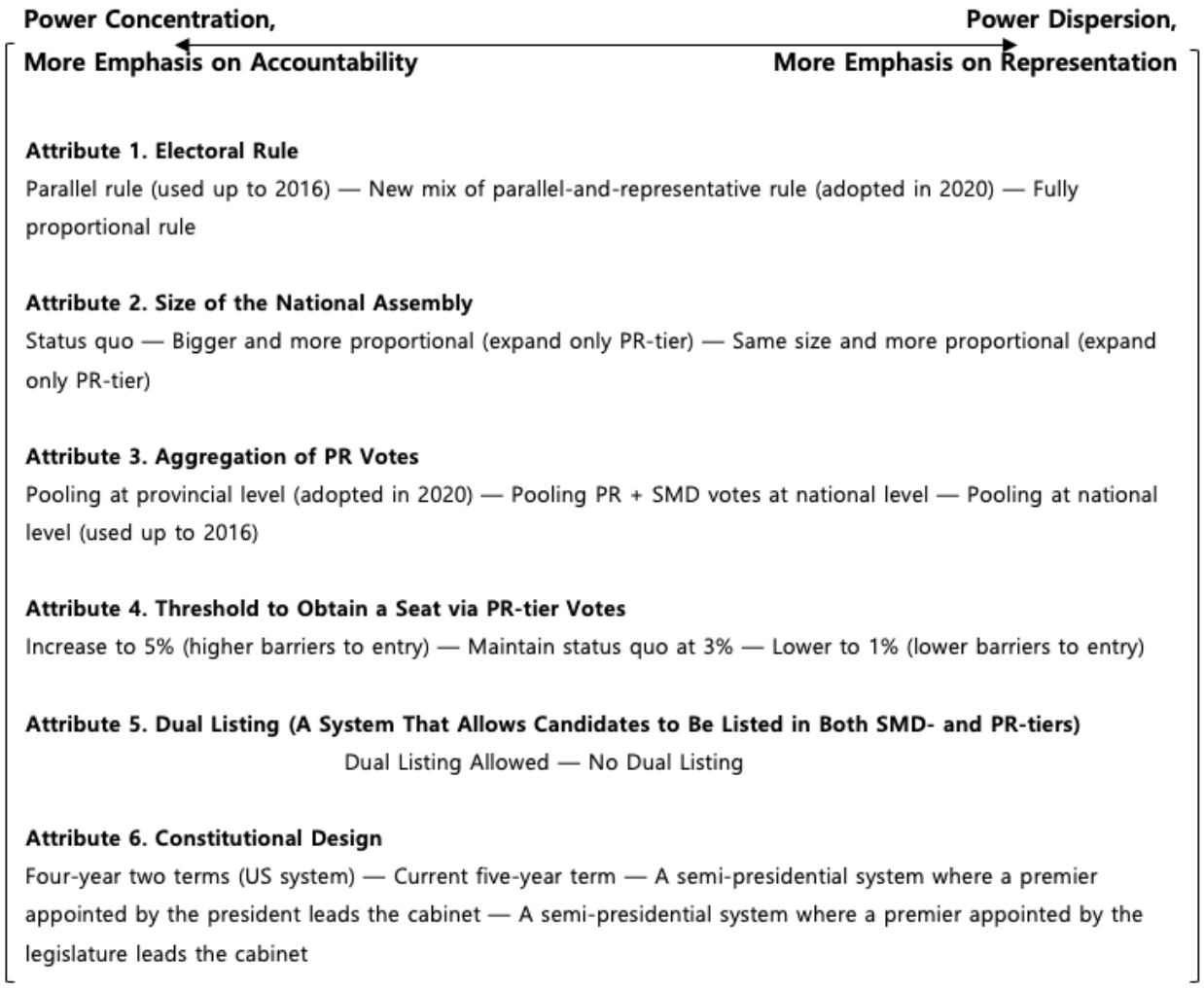
	Age of Respondents					
	Under 40		40 or Older			
	Mean	SD	Mean	SD	Difference	SE
Female	0.46	(0.50)	0.40	(0.49)	0.05*	(0.03)
Party Affiliation						
Democratic Party	0.27	(0.44)	0.32	(0.47)	-0.05**	(0.02)
Mirae Party	0.11	(0.31)	0.11	(0.31)	0.00	(0.02)
Minor Party or Independent	0.62	(0.49)	0.57	(0.49)	0.05*	(0.02)
N	514		1,152			

2 Conjoint Experiment

Question Wording Translated in English

Recently, the National Assembly implemented an electoral reform from the existing parallel election system centered on single-member districts to the mix of parallel and proportional representation systems that strengthens proportionality and representation. However, decisions on details such as expansion of a PR-tier and expansion of the size of the National Assembly can be re-discussed after the general election. If your opinion was put to a vote in this general election, what choice would you make? Before answering the questions, we present the following table to help you understand the characteristics of institutional rules. Among the details belonging to each element, the more it is placed on the left, the more emphasis it is on the accountability aspect (concentrating power on a few major parties), and the more it is placed on the right, the more emphasis it is on the representation aspect (reflecting the opinions of diverse parties).

Figure A1: Information Table (English Version)



Notes: Attributes were shown in a randomized order at the respondent level. Levels of each attribute were randomly assigned based on the values in Table 1. Each respondent was asked to answer up to five times, with a new set of attribute values for each option being presented through randomization each time.

Based on this, when you are given a choice between the following two systems, which one do you prefer?

Table A3: Conjoint Experiment Design (English Version)

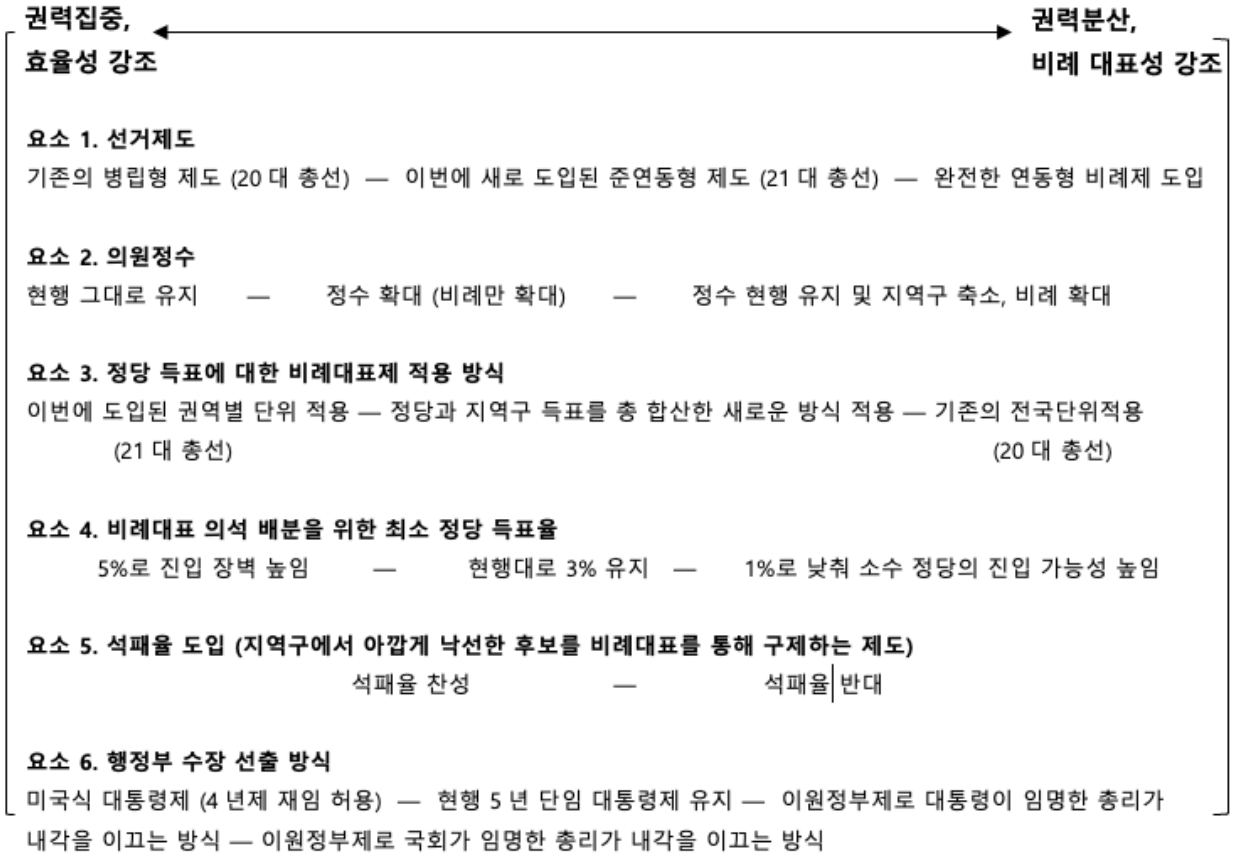
	Option A	**Option B**
Electoral Rule		
Size of the National Assembly		
Aggregation of PR Votes		
Threshold to Obtain a PR Seat		
Dual Listing		
Constitutional Design		

Original Korean Version

최근, 국회에서는 소선거구 다수대표제를 중심으로 한 기존의 병립형 선거제도로부터 비례성과 대표성을 강화하는 준연동형 비례대표제로의 선거제도 개혁을 실시하였습니다. 하지만, 비례대표 확대, 의원 정수 확대 등과 같은 세부 사항에 대한 결정은 이번 총선 이후 재 논의될 가능성이 있습니다. 만약, 이번 총선에서 이에 대해 국민 여러분의 의견이 투표에 부쳐진다면 어떠한 선택을 하시겠습니까?

질문에 응답하시기에 앞서서 여러분의 이해를 돕고자 다음과 같은 표를 준비했습니다. 각 요소에 속한 세부 사항 중 왼쪽에 위치할수록 기존 주요 정당에 권력이 집중되기 유리한 제도이고 (효율성 강조), 오른쪽에 위치할수록 다양한 정당의 의견이 좀 더 반영될 수 있는 제도입니다 (대표성 강조).

Figure A2: Information Table (Korean Version)



이를 바탕으로 다음과 같은 두 가지 제도에 대해 선택이 주어질 때, 어떠한 쪽을 선호하십니까?

Table A4: Conjoint Experiment Design (Korean Version)

	선택 A	**선택 B**
선거 제도		
의원 정수		
정당 득표에 대한 비례대표제 적용 방식		
비례대표 의석 배분을 위한 최소 정당 득표율		
석패율 도입		
행정부 수장 선출 방식		

A Note on Terminology

In designing the Korean-language survey instrument, we selected terms that would most clearly convey our theoretical interest in the trade-off between representation and accountability. For "representation," we used the standard Korean term "대표성," which is widely

used in media coverage, academic writing, and legislative debates to describe the degree to which electoral outcomes reflect the preferences and composition of diverse groups.

For the accountability side of the trade-off, we intentionally used "efficiency (효율성)" rather than a literal translation of "accountability (책무성)." In Korean, the direct translation of "accountability" typically conveys administrative responsibility (책임성), which does not fully capture our intended meaning: voters' ability to identify governing alternatives and hold them to account through elections. In discussions of electoral systems in Korean, "효율성" is the more familiar term and used to describe this feature of majoritarian rules. Although efficiency and accountability are not identical concepts, in this specific context "효율성" is the term that best approximates the accountability mechanism we aim to capture.

To illustrate how these concepts appear in public discussions of electoral reform, we provide an excerpt from an op-ed in one of Korean's major newspapers, the *JoongAng Ilbo*, which highlights the trade-offs between majoritarian and proportional rules.

Korean original:

"선거구제 개편은 민주정치를 어떻게 운영하는 것이 순리인가를 숙고하면서 구상해야 한다. 양당 경쟁에서 승리한 정당에 권력을 집중시켜 국정운영의 **효율성** 및 책임성을 확보해야 한다는 논리가 있고, 다당제 맥락에서 의회의 **대표성**을 확대하고 폭넓은 합의를 형성해야 한다는 논리도 있다. 소선거구 상대다수제가 전자에, 중 · 대선거구 비례대표제가 후자에 부합한다. 한국의 정치제도는 전반적으로 권력 집중 성격이 강하여 후자의 논리가 좀 더 강화될 필요가 있다. 그렇지만 **대표성** 강화와 합의 형성을 위한 제도 변경은 현실 여건과 발생될 비용을 고려하는 균형적인 관점에서 추진해야 한다. " (emphasis added)

English translation:

"Electoral reform should be conceived with careful consideration of how to operate democratic politics. Some argue that power should be concentrated in the party that wins the two-party race to ensure **efficiency (accountability)** and responsibility in state administration. Others argue that, within a multi-party system, parliamentary **representation** should be expanded and broad consensus should be fostered. The single-member district ma-

joritarian system fits the former, while the medium- and large-member district proportional representation system fits the latter. Given the overall centralization of power in Korea's political system, the latter argument needs to be strengthened. However, institutional changes aimed at strengthening **representation** and consensus building should be pursued with a balanced perspective, taking into account current circumstances and the potential costs." (emphasis added)

Citation: Park Chan-wook, "선거구제 개편, 현 2표 병립제의 장점은 살려야 (Electoral Reform: Preserving the Strengths of the Current Two-Member District System)," *JoongAng Ilbo*, June 22, 2010, https://www.joongang.co.kr/article/4261850 (accessed November 2025).

3 Results

Table A5: Electoral System Preferences

	Estimated AMCE	Standard Error	N
Major Rule			
Electoral System			
MMP	0.011	(0.010)	16,562
Mix of MMM and MMP	0.004	(0.010)	16,562
MMM	–	–	16,562
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	0.007	(0.010)	16,562
Bigger Size, More Proportional	-0.036***	(0.010)	16,562
Status Quo	–	–	16,562
Aggregation of PR Votes			
PR Pools at National Level	–	–	16,562
PR and SMD Votes Pool at National Level	-0.001	(0.009)	16,562
PR Pools at Provincial Level	-0.016*	(0.009)	16,562
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.004	(0.010)	16,562
3 Percent of PR Vote	–	–	16,562
5 Percent of PR Vote	-0.014	(0.010)	16,562
Dual Listing			
No Dual Listing	–	–	16,562
Dual Listing	-0.011	(0.008)	16,562

Notes: Table version of Figure 1. Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable in our conjoint analysis is equal to 1 if a profile is chosen and 0 otherwise. *p<.1; **p<.05; ***p<.01.

Table A6: Results by Gender

	Female	Male	Female–Male
Major Rule			
Electoral System			
MMP	0.009	0.014	-0.004
	(0.015)	(0.012)	(0.019)
Mix of MMM and MMP	0.045***	-0.026**	0.071***
	(0.016)	(0.013)	(0.020)
MMM	–	–	–
	–	–	–
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	0.024*	-0.006	0.030
	(0.015)	(0.013)	(0.020)
Bigger Size, More Proportional	-0.017	-0.051***	0.033*
	(0.015)	(0.013)	(0.020)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	-0.003	0.0004	-0.004
	(0.015)	(0.013)	(0.019)
PR Pools at Provincial Level	-0.010	-0.022*	0.012
	(0.014)	(0.013)	(0.019)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.036**	-0.019	0.055***
	(0.015)	(0.013)	(0.020)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	0.010	-0.032**	0.042**
	(0.015)	(0.013)	(0.020)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.005	-0.015	0.009
	(0.013)	(0.011)	(0.017)
N	6,950	9,612	16,562

Notes: Table version of Figure 2. Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A7: Results by Age

	Under 40	40 or Older	Under 40–40 or Older
Major Rule			
Electoral System			
MMP	0.046**	-0.004	0.049**
	(0.017)	(0.011)	(0.021)
Mix of MMM and MMP	0.008	0.003	0.006
	(0.018)	(0.012)	(0.022)
MMM	–	–	–
	–	–	–
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	-0.013	0.016	-0.029
	(0.017)	(0.012)	(0.020)
Bigger Size, More Proportional	-0.024	-0.042***	0.018
	(0.018)	(0.012)	(0.021)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	0.014	-0.008	0.022
	(0.017)	(0.011)	(0.020)
PR Pools at Provincial Level	-0.021	-0.014	-0.007
	(0.017)	(0.012)	(0.020)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.018	-0.002	0.020
	(0.017)	(0.012)	(0.021)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	-0.021	-0.010	-0.011
	(0.017)	(0.012)	(0.021)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.019	-0.008	-0.012
	(0.015)	(0.010)	(0.018)
N	5,098	11,464	16,562

Notes: Table version of Figure 3. Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A8: Results by Age Group

	18–29	30–39	40–49	50–59	60 or Older
Major Rule					
Electoral System					
MMP	0.020	0.066***	-0.013	0.014	-0.015
	(0.025)	(0.024)	(0.018)	(0.019)	(0.024)
Mix of MMM and MMP	0.036	-0.014	0.013	0.017	-0.039*
	(0.027)	(0.024)	(0.019)	(0.020)	(0.023)
MMM	–	–	–	–	–
	–	–	–	–	–
Minor Rules					
Size of the National Assembly					
Same Size, More Proportional	-0.006	-0.018	0.017	0.039**	-0.021
	(0.025)	(0.023)	(0.019)	(0.019)	(0.024)
Bigger Size, More Proportional	-0.008	-0.038	-0.022	-0.028	-0.094***
	(0.026)	(0.024)	(0.018)	(0.020)	(0.024)
Status Quo	–	–	–	–	–
	–	–	–	–	–
Aggregation of PR Votes					
PR Pools at National Level	–	–	–	–	–
	–	–	–	–	–
PR and SMD Votes Pool at National Level	0.024	0.007	-0.007	-0.004	-0.016
	(0.025)	(0.023)	(0.018)	(0.019)	(0.025)
PR Pools at Provincial Level	-0.018	-0.023	-0.001	0.004	-0.067***
	(0.026)	(0.022)	(0.019)	(0.019)	(0.022)
Threshold to Obtain a PR Seat					
1 Percent of PR Vote	0.016	0.018	0.002	-0.007	-0.003
	(0.026)	(0.023)	(0.020)	(0.018)	(0.025)
3 Percent of PR Vote	–	–	–	–	–
	–	–	–	–	–
5 Percent of PR Vote	0.005	-0.044*	-0.006	-0.024	0.005
	(0.025)	(0.024)	(0.019)	(0.019)	(0.024)
Dual Listing					
No Dual Listing	–	–	–	–	–
	–	–	–	–	–
Dual Listing	0.007	-0.040**	0.009	0.016	-0.075***
	(0.023)	(0.020)	(0.016)	(0.017)	(0.020)
N	2,306	2,792	4,504	4,200	2,760

Notes: Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<.1; **p<.05; ***p<.01.

Table A9: Results by Gender and Age

	Female Under 40	Female 40 or Older	Male Under 40	Male 40 or Older
Major Rule				
Electoral System				
MMP	0.047*	-0.009	0.045*	0.001
	(0.026)	(0.018)	(0.023)	(0.015)
Mix of MMM and MMP	0.056**	0.040**	-0.032	-0.024
	(0.027)	(0.019)	(0.024)	(0.015)
MMM	–	–	–	–
	–	–	–	–
Minor Rules				
Size of the National Assembly				
Same Size, More Proportional	0.008	0.030*	-0.032	0.005
	(0.025)	(0.018)	(0.023)	(0.016)
Bigger Size, More Proportional	0.003	-0.027	-0.049**	-0.052***
	(0.026)	(0.018)	(0.024)	(0.015)
Status Quo	–	–	–	–
	–	–	–	–
Aggregation of PR Votes				
PR Pools at National Level	–	–	–	–
	–	–	–	–
PR and SMD Votes Pool at National Level	0.013	-0.012	0.014	-0.005
	(0.025)	(0.018)	(0.023)	(0.015)
PR Pools at Provincial Level	-0.017	-0.007	-0.026	-0.020
	(0.024)	(0.018)	(0.023)	(0.015)
Threshold to Obtain a PR Seat				
1 Percent of PR Vote	0.039	0.034*	-0.0002	-0.027*
	(0.026)	(0.018)	(0.023)	(0.016)
3 Percent of PR Vote	–	–	–	–
	–	–	–	–
5 Percent of PR Vote	-0.009	0.020	-0.032	-0.031**
	(0.026)	(0.018)	(0.023)	(0.016)
Dual Listing				
No Dual Listing	–	–	–	–
	–	–	–	–
Dual Listing	-0.022	0.002	-0.017	-0.015
	(0.023)	(0.016)	(0.020)	(0.013)
N	2,314	4,636	2,784	6,828

Notes: Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<.1; **p<.05; ***p<.01.

Table A10: Results by Party

	Democratic Party	Mirae Party	Minor Party or Independent
Major Rule			
Electoral System			
MMP	0.010	-0.019	0.018
	(0.017)	(0.028)	(0.012)
Mix of MMM and MMP	0.029	-0.039	-0.001
	(0.019)	(0.029)	(0.013)
MMM	–	–	–
	–	–	–
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	0.019	-0.032	0.008
	(0.017)	(0.029)	(0.013)
Bigger Size, More Proportional	-0.026	-0.070**	-0.036***
	(0.018)	(0.031)	(0.013)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	0.015	-0.014	-0.007
	(0.017)	(0.029)	(0.012)
PR Pools at Provincial Level	0.039**	-0.074***	-0.035***
	(0.017)	(0.027)	(0.013)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.001	-0.032	0.012
	(0.018)	(0.028)	(0.013)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	0.002	-0.084***	-0.009
	(0.018)	(0.028)	(0.013)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.017	-0.016	-0.007
	(0.015)	(0.024)	(0.001)
N	5,008	1,822	9,732

Notes: Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A11: Results by Gender and Party

	Democratic Party	Mirae Party	Minor Party or Independent
Major Rule			
Electoral System			
MMP	0.035	-0.099	-0.013
	(0.034)	(0.062)	(0.025)
Mix of MMM and MMP	0.118***	0.010	0.055**
	(0.038)	(0.059)	(0.026)
MMM	–	–	–
	–	–	–
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	0.007	0.105*	0.026
	(0.035)	(0.061)	(0.026)
Bigger Size, More Proportional	-0.006	-0.053	0.067***
	(0.035)	(0.064)	(0.026)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	0.034	0.015	-0.027
	(0.034)	(0.064)	(0.025)
PR Pools at Provincial Level	0.017	0.045	0.0003
	(0.034)	(0.058)	(0.025)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.029	0.060	0.067**
	(0.035)	(0.061)	(0.026)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	0.068*	0.050	0.022
	(0.036)	(0.060)	(0.025)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.010	-0.016	0.023
	(0.030)	(0.053)	(0.023)
N	5,008	1,822	9,732

Notes: Table version of Figure 5. Table entries are estimated differences in average marginal component effects (AMCEs) between women and men with clustered standard errors at the respondent level. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A12: Results by Age and Party

	Democratic Party	Mirae Party	Minor Party or Independent
Major Rule			
Electoral System			
MMP	0.053	0.008	0.056**
	(0.039)	(0.059)	(0.027)
Mix of MMM and MMP	0.005	-0.026	0.014
	(0.042)	(0.066)	(0.027)
MMM	–	–	–
	–	–	–
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	-0.076**	0.027	-0.020
	(0.038)	(0.062)	(0.027)
Bigger Size, More Proportional	0.018	0.033	0.012
	(0.039)	(0.072)	(0.027)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	0.032	0.099*	0.007
	(0.037)	(0.057)	(0.027)
PR Pools at Provincial Level	-0.010	0.026	-0.004
	(0.036)	(0.056)	(0.027)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	-0.050	0.032	0.050*
	(0.040)	(0.061)	(0.027)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	0.016	-0.032	-0.016
	(0.042)	(0.060)	(0.026)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.041	0.012	-0.002
	(0.033)	(0.053)	(0.024)
N	5,008	1,822	9,732

Notes: Table version of Figure 6. Table entries are estimated differences in average marginal component effects (AMCEs) between respondents under 40 and those 40 or older with clustered standard errors at the respondent level. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A13: Results by Gender, Age, and Party

	Democratic Party	Mirae Party	Minor Party or Independent
Major Rule			
Electoral System			
MMP	0.059	-0.035	0.047
	(0.054)	(0.103)	(0.038)
Mix of MMM and MMP	0.133**	-0.063	0.074**
	(0.058)	(0.123)	(0.039)
MMM	–	–	–
	–	–	–
Minor Rules			
Size of the National Assembly			
Same Size, More Proportional	-0.065	0.089	0.008
	(0.055)	(0.100)	(0.037)
Bigger Size, More Proportional	0.011	-0.057	0.081**
	(0.057)	(0.123)	(0.038)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	0.051	0.121	-0.011
	(0.057)	(0.037)	(0.081)
PR Pools at Provincial Level	-0.013	0.079	0.004
	(0.055)	(0.036)	(0.120)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	-0.014	0.045	0.104***
	(0.058)	(0.128)	(0.038)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	0.063	0.044	0.004
	(0.062)	(0.094)	(0.037)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.067	0.024	0.013
	(0.047)	(0.080)	(0.034)
N	2,688	984	5,470

Notes: Table version of Figure 7. Table entries are estimated differences in average marginal component effects (AMCEs) between women under 40 and men 40 or older with clustered standard errors at the respondent level. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A14: Results by Income

	Lower Income	Higher Income	Lower–Higher
Electoral Rule			
MMP	0.020	0.00004	0.020
	(0.013)	(0.014)	(0.019)
Mix of MMM and MMP	0.014	-0.007	0.021
	(0.013)	(0.015)	(0.020)
MMM	–	–	–
	–	–	–
Size of the National Assembly			
Same Size, More Proportional	0.024*	-0.013	0.036*
	(0.013)	(0.015)	(0.019)
Bigger Size, More Proportional	-0.021*	-0.056***	0.035*
	(0.013)	(0.015)	(0.020)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	-0.016	0.019	-0.036*
	(0.012)	(0.015)	(0.019)
PR Pools at Provincial Level	-0.022	-0.010	-0.012
	(0.013)	(0.014)	(0.019)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.009	-0.004	0.013
	(0.013)	(0.015)	(0.020)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	-0.009	-0.018	0.009
	(0.013)	(0.015)	(0.020)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.011	-0.013	0.002
	(0.011)	(0.013)	(0.017)
Constitutional Design			
Semi-Presidential (Premier-Presidential)	-0.106***	-0.074***	-0.032
	(0.016)	(0.019)	(0.025)
Semi-Presidential (President-Parliamentary)	-0.080***	-0.051***	-0.036
	(0.016)	(0.018)	(0.024)
Current System (Five Year, One Term)	–	–	–
	–	–	–
US Style (Four Year, Two Terms)	0.004	0.034*	-0.030
	(0.017)	(0.019)	(0.026)
N	9,444	7,118	16,562

Notes: Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A15: Results by Approval of President Moon Jae-in

	Approve Moon	Disapprove Moon	Approve–Disapprove
Electoral Rule			
MMP	0.028**	-0.004	0.032*
	(0.014)	(0.013)	(0.019)
Mix of MMM and MMP	0.030**	-0.018	0.049**
	(0.015)	(0.013)	(0.020)
MMM	–	–	–
	–	–	–
Size of the National Assembly			
Same Size, More Proportional	0.033**	-0.016	0.049**
	(0.014)	(0.013)	(0.019)
Bigger Size, More Proportional	-0.009	-0.060***	0.051***
	(0.014)	(0.013)	(0.019)
Status Quo	–	–	–
	–	–	–
Aggregation of PR Votes			
PR Pools at National Level	–	–	–
	–	–	–
PR and SMD Votes Pool at National Level	0.005	-0.004	0.009
	(0.014)	(0.013)	(0.019)
PR Pools at Provincial Level	0.017	-0.048***	0.065***
	(0.014)	(0.013)	(0.019)
Threshold to Obtain a PR Seat			
1 Percent of PR Vote	0.003	0.004	-0.001
	(0.014)	(0.014)	(0.020)
3 Percent of PR Vote	–	–	–
	–	–	–
5 Percent of PR Vote	-0.020	-0.006	-0.014
	(0.014)	(0.014)	(0.019)
Dual Listing			
No Dual Listing	–	–	–
	–	–	–
Dual Listing	-0.015	-0.011	-0.004
	(0.012)	(0.012)	(0.017)
Constitutional Design			
Semi-Presidential (Premier-Presidential)	-0.090***	-0.093***	0.003
	(0.018)	(0.017)	(0.025)
Semi-Presidential (President-Parliamentary)	-0.050***	-0.091***	0.040*
	(0.017)	(0.016)	(0.023)
Current System (Five Year, One Term)	–	–	–
	–	–	–
US Style (Four Year, Two Terms)	0.078***	-0.041**	0.119***
	(0.018)	(0.018)	(0.026)
N	7,968	8,594	16,562

Notes: Table entries are estimated AMCEs (Average Marginal Component Effects) with standard errors clustered by respondent in parentheses. For each attribute, we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable is equal to 1 if a profile is chosen and 0 otherwise. *p<0.1; **p<0.05; ***p<0.01.

Table A16: Executive Branch Preferences

	Estimated AMCE	Standard Error	N
Constitutional Design			
Semi-Presidential (Premier-Presidential)	-0.092***	(0.012)	16,562
Semi-Presidential (President-Parliamentary)	-0.072***	(0.012)	16,562
Current System (Five Year, One Term)	–	–	16,562
US Style (Four Year, Two Terms)	0.017	(0.013)	16,562

Notes: While we do not focus on executive branch preferences in this article, including this variable in our other conjoint analyses has no effect on the substantive results. Table entries are average marginal component effect (AMCE) estimates with standard errors clustered by respondent in parentheses. As with other attributes (Table A5), we set the baseline to the status quo and organize the levels such that they range from the one most emphasizing representation at the top to the one most prioritizing accountability at the bottom. The dependent variable in our conjoint analysis is equal to 1 if a profile is chosen and 0 otherwise. *p<.1; **p<.05; ***p<.01.

4 Ethical Issues and Considerations

The research in this paper fully conforms with the Principles and Guidelines for Human Subjects Research as approved by the APSA Council on April 4, 2020.

In working with Research & Research, a major Korean survey firm, we made sure to consider the well-being of participants at every step in the research process. We received approval for our conjoint experiment from the Institutional Review Board at an accredited university. In our survey, we provided a brief description beforehand and then only admitted respondents who said yes to participating our survey. We told respondents that they were free to stop answering our survey at any time, and we gave a debriefing statement at the end of the survey. Our survey involved no deception and only minimal likelihood of harm. We did present respondents with fictitious profiles of electoral rules, but we clearly informed participants that these were only hypothetical sets of electoral rules.

Participants in our survey came from the subject pool of Research & Research. Respondents were paid fairly for their time at the standard rate in South Korea for online survey participants: 10,000 won (approximately $7.40). The participant pool was diverse and we specifically selected respondents to match the latest census in terms of age, sex, and region of residence. We only recruited individuals of eligible voting age in South Korea (18 and above). Our participant pool thus targeted regular, volunteer participants in Research & Research surveys rather than any specific marginalized or vulnerable groups. Research & Research assured us that the confidentiality and anonymity of our participants would be protected, and we repeatedly informed participants that there was no penalty for not completing our survey. Finally, we made sure that our study would not violate any relevant laws and regulations governing research and related activities.

5 Pre-Analysis Plan

This study was pre-registered at Open Science Framework (OSF) before we accessed the data and began our analysis. We have copied the submitted pre-analysis plan (PAP) below, with minor edits for formatting.

Title of Project

Electoral Reform and Voter Preferences: Evidence from a Conjoint Experiment

Authorship

Anonymous author(s)

Brief Summary of Project

The objective of this project is to understand whether more representative electoral rules induce greater support from marginalized voters in society, as intended by lawmakers. There is a debate in the current literature on electoral reform about the trade-off between accountability and representation that exists in majoritarian versus representative rules. Yet, there is little research on whether different types of voter groups prefer different electoral rules. We conduct a conjoint experiment leveraging an actual reform in South Korea, which changed the electoral system to be less majoritarian and more representative. We obtain over 8,000 responses from a nationally representative survey, conducted immediately before Korea's 2020 general election, to tease out whether voter preferences concerning electoral rules vary across diverse individual characteristics, including demographic, socioeconomic, and political characteristics. Our experimental research has important implications for electoral reform, voter preferences, and the representation of marginalized groups.

Study Design

Hypotheses and How Treatment Effects Will Be Measured

First, we expect that marginalized voters, such as women and younger people, are more likely to support political reforms that adopt more representative electoral systems and constitutional designs than dominant voter groups, such as men and older people.

Second, we also test whether voters of lower socioeconomic status tend to be more supportive of representative electoral systems and constitutional designs than those of higher socioeconomic status.

In addition, given that the ruling left-wing party was more supportive of the political reform than the opposition right-wing party, we test whether voters who approve of President Moon are more likely to favor representative electoral systems than those who disapprove of President Moon.

Main Variables of Interest and Measurements of Treatment Effects

Since our analysis is based on a conjoint experiment, our dependent variable is a choice out of two sets of features that show a combination of different components of electoral rules and constitutional designs that will be randomly presented.

Our independent variables are individual characteristics of voters, including demographic (gender and age), socioeconomic (income, occupation), and political (party ID) characteristics.

In terms of measuring the treatment effects, we plan to analyze the data from our conjoint experiment in two ways. First, we will look at descriptive marginal means (MMs), which show the mean outcome across all appearances of a particular conjoint feature, averaging across all other features. Second, we will also follow the more common approach in conjoint analyses which is to calculate and show the average marginal component effects (AMCEs) using some form of regression analysis.

Distinction Between Primary and Secondary Outcomes

While our analysis has no clear distinction between primary and secondary outcomes, our main analysis is related to testing whether marginalized groups are more supportive of political reforms that adopt more representative electoral systems and constitutional designs than dominant voter groups in society. In addition, we also aim to analyze the effects of socioeconomic and political characteristics on voter preferences concerning political reforms to test our predictions described above.

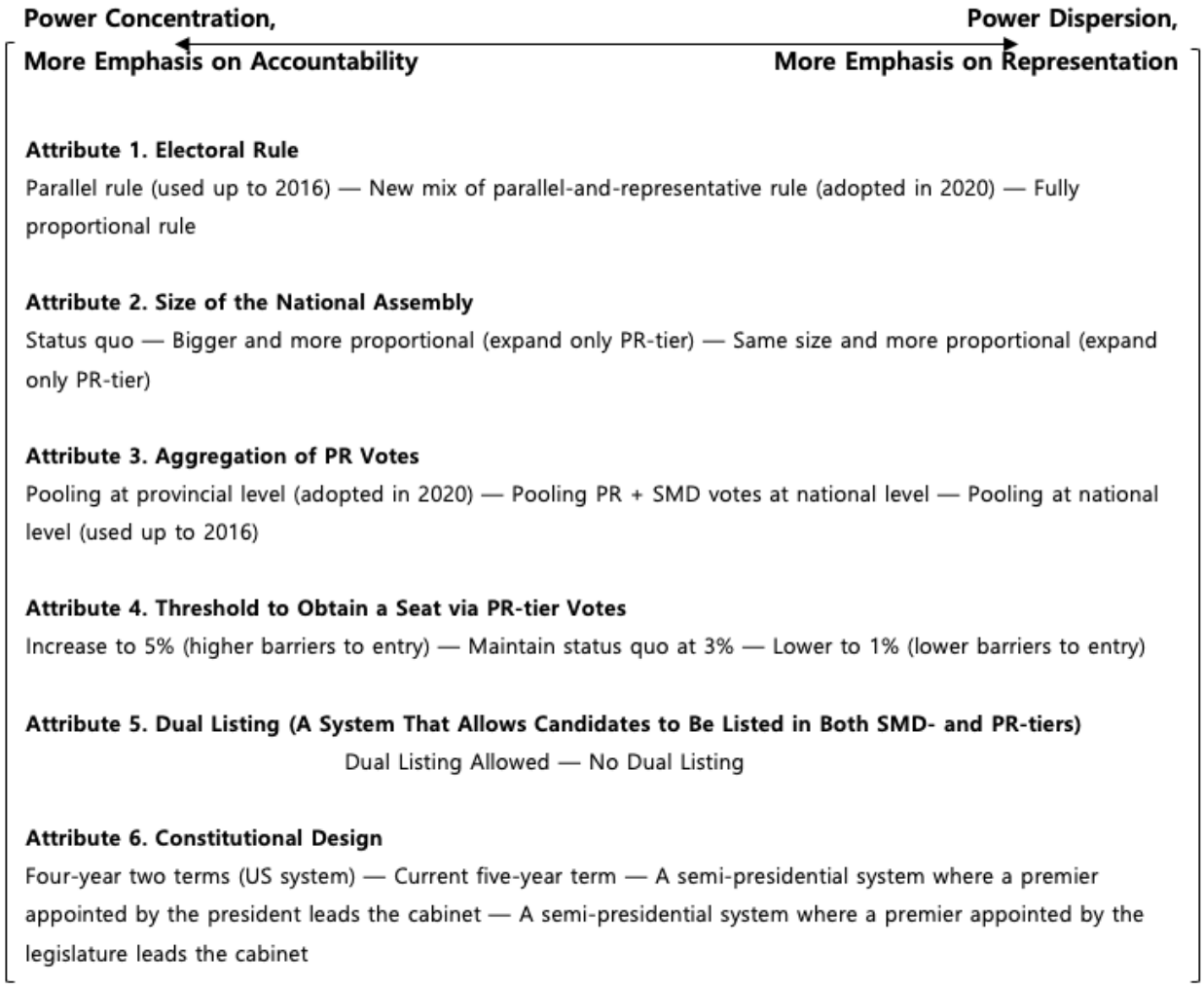
Details of Study

We conducted our conjoint survey experiment with a nationally representative sample of more than 1,800 South Korean adults who are eligible voters from late March to early April 2020, which is before the legislative election that took place on April 15, 2020. The survey sample was recruited by Research & Research, one of the largest survey research firms in South Korea. The survey design and administration was conducted using Qualtrics, and Research & Research ensured the recruitment of a nationally representative sample and the overall seamless implementation of the survey. The human research subjects aspect of our experimental protocol was approved by our university's Institutional Review Board. Below, our actual conjoint survey experiment wording is described.

Recently, the National Assembly implemented an electoral reform from the existing parallel election system centered on single-member districts to the mix of parallel and proportional representation systems that strengthens proportionality and representation. However, decisions on details such as expansion of a PR-tier and expansion of the size of the National Assembly can be re-discussed after the general election. If your opinion was put to a vote in this general election, what choice would you make? Before answering the questions, we present the following table to help you understand the characteristics of institutional rules. Among the details belonging to each element, the more it is placed on the left, the more advantageous it is to concentrating power on large political parties, and the more it is placed

on the right, the more the opinions of minority parties can be reflected.

Figure A3: Information Table (English Version)



Based on this, when you are given a choice between the following two systems, which one do you prefer?

Table A17: Conjoint Experiment Design (English Version)

	Option A	**Option B**
Electoral Rule		
Size of the National Assembly		
Aggregation of PR Votes		
Threshold to Obtain a PR Seat		
Dual Listing		
Constitutional Design		

Notes: Attributes were shown in a randomized order at the respondent level. Levels of each attribute were randomly assigned based on the values in Table 1. Each respondent was asked to answer up to five times, with a new set of attribute values for each option being presented through randomization each time.